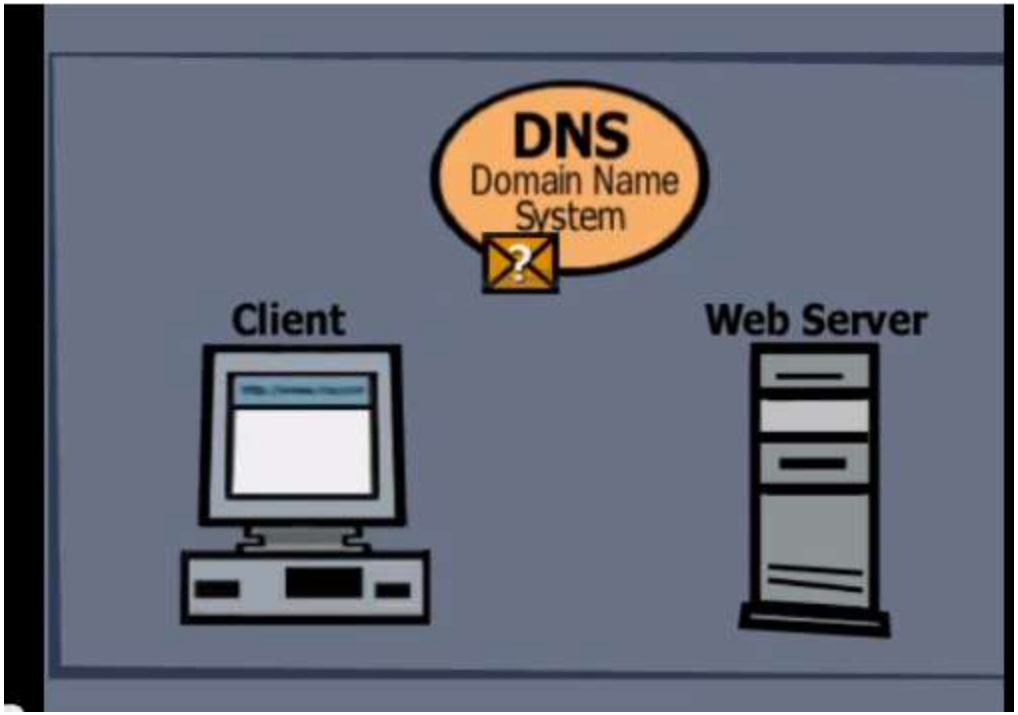
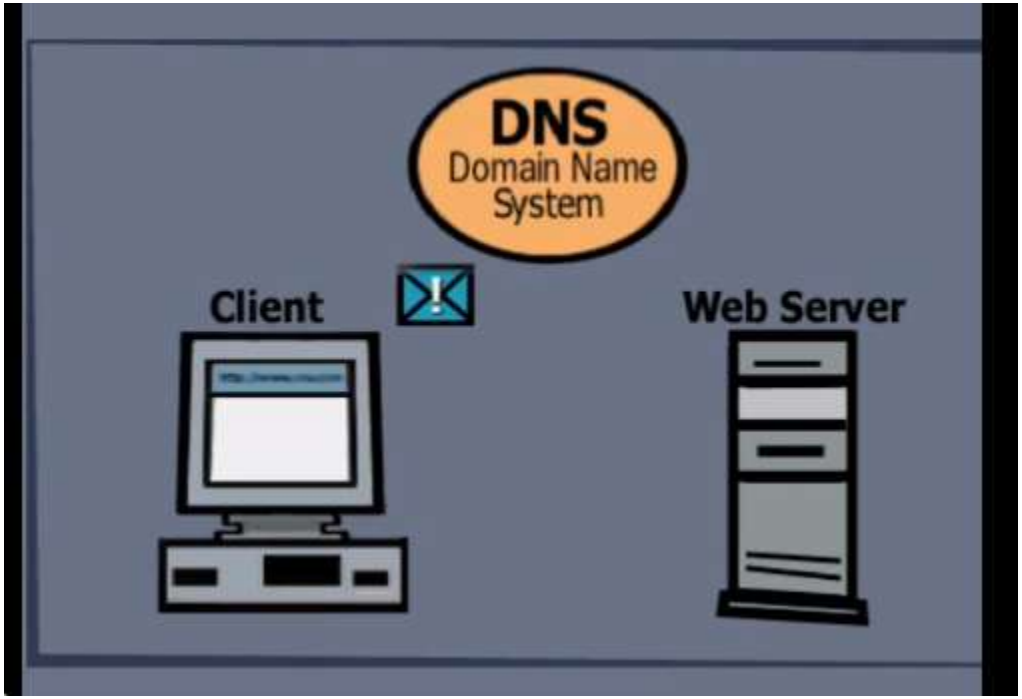
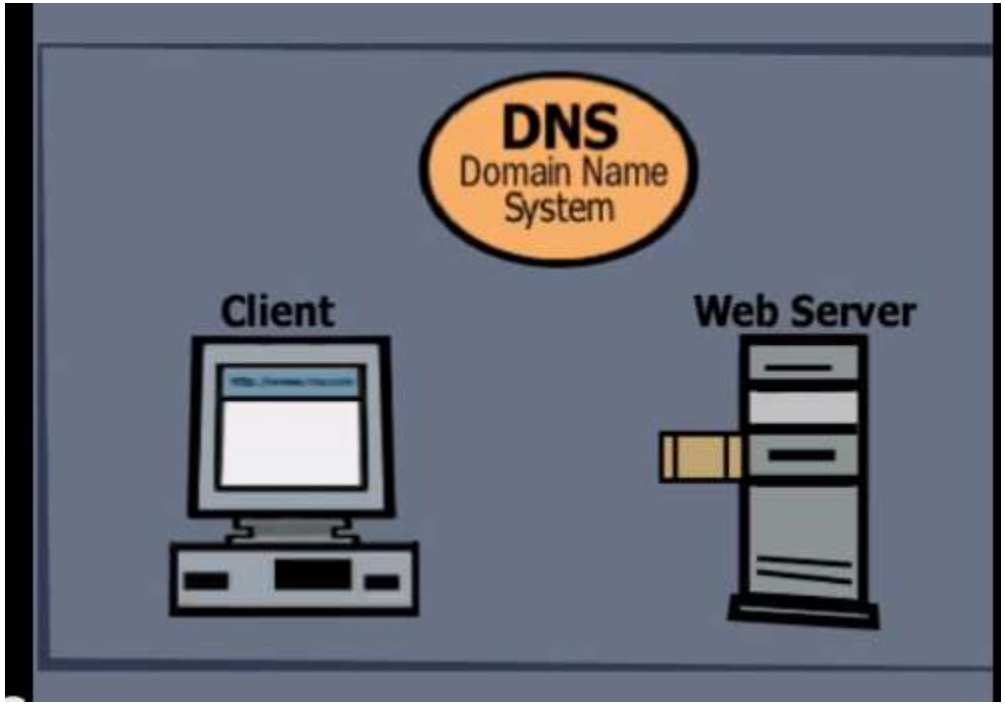
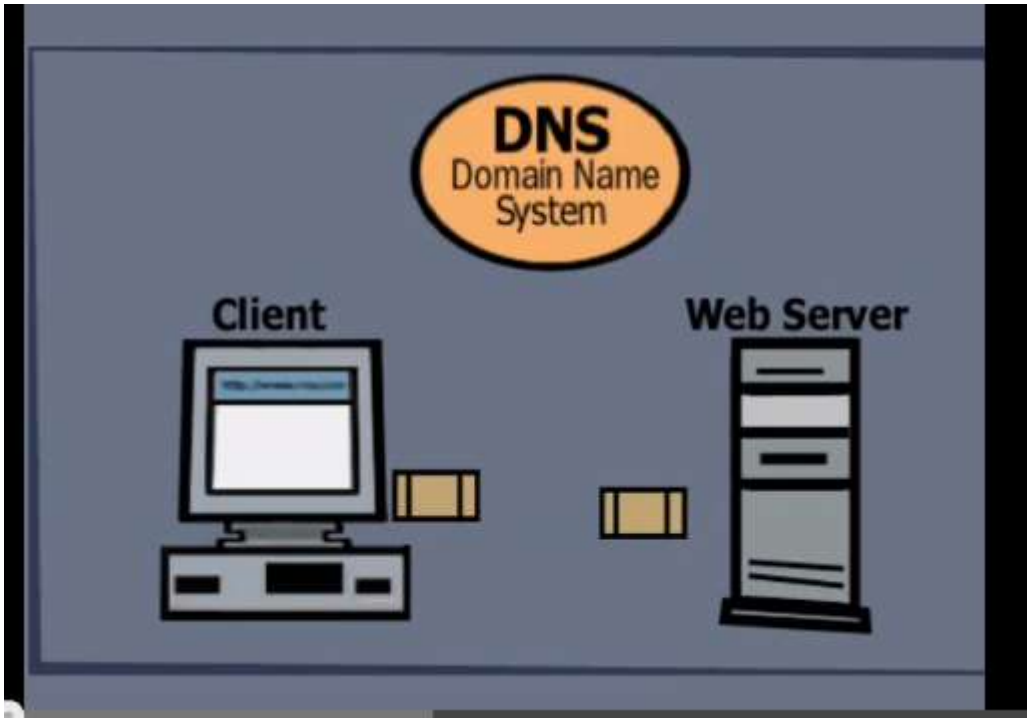
- Introduction to DNS

The Domain Name System is a distributed database system that can serve as the foundation for name resolution in a TCP/IP network.


DNS
Domain Name
System
?
Client
Web Server


DNS
Domain Name
System
Client
Web Server


DNS
Domain Name
System
Client
Web Server


DNS
Domain Name
System
Client
Web Server

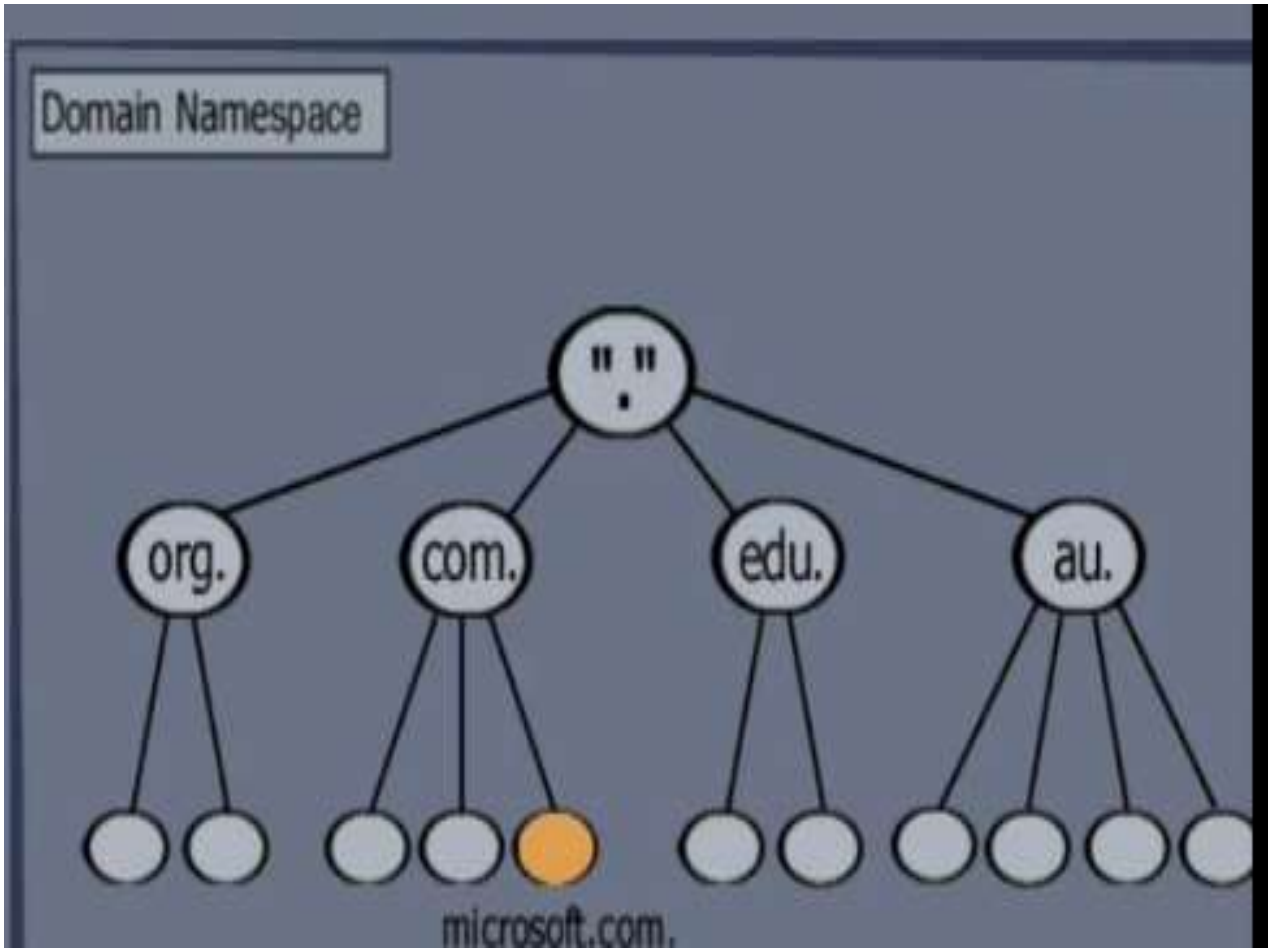
DNS Benefits

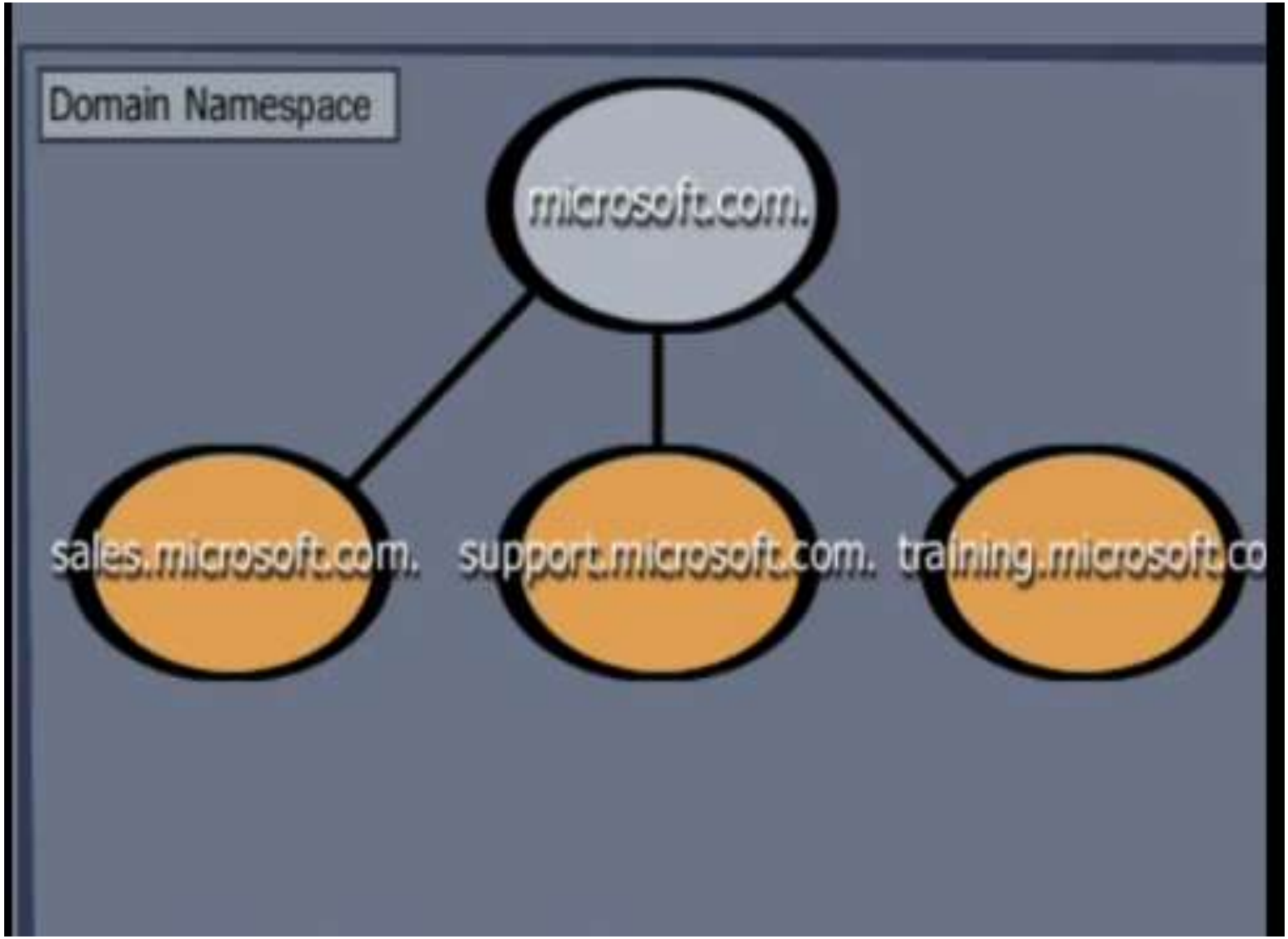
Convenience
names are easier
to remember

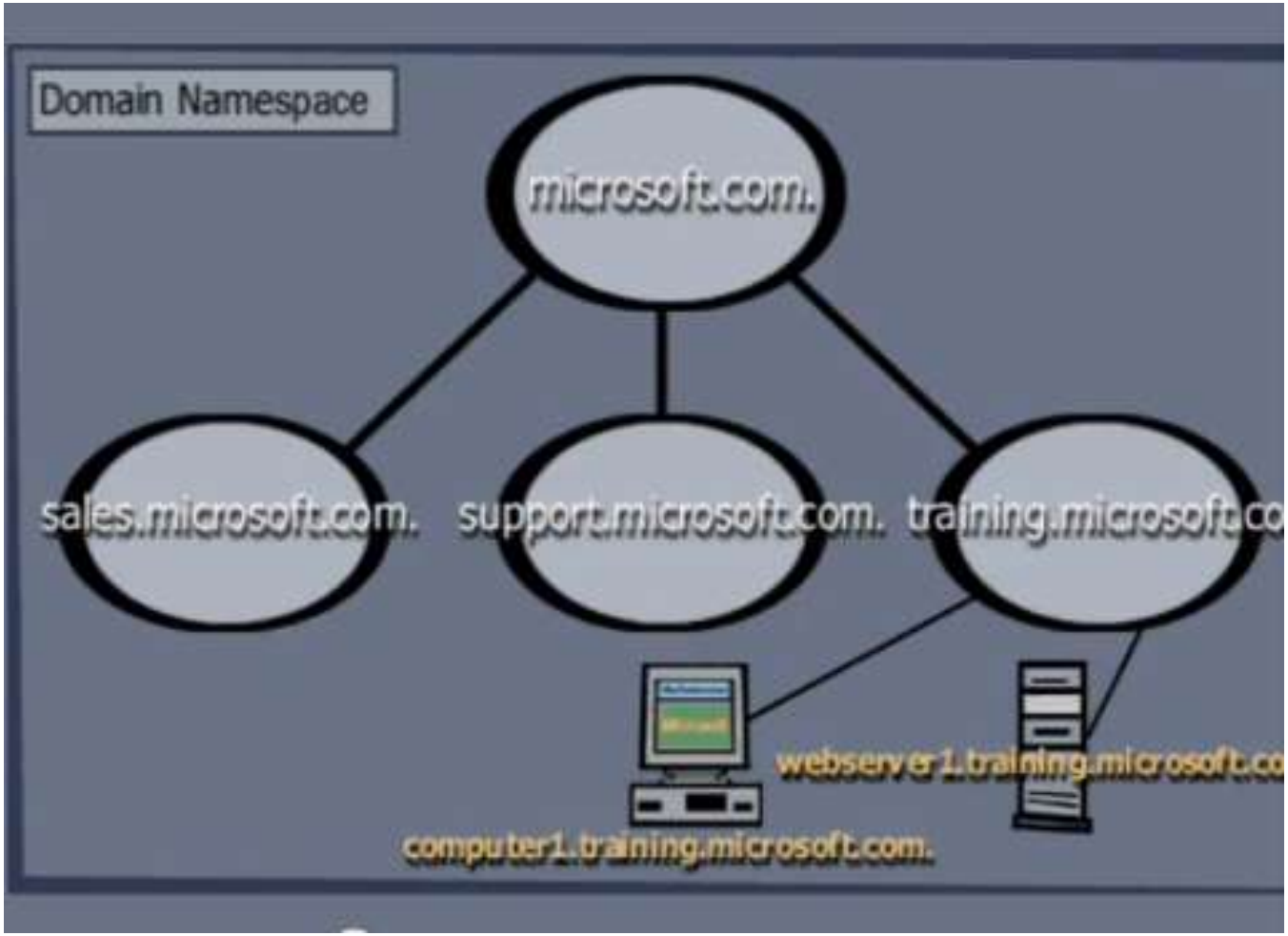
Simplicity
one naming
convention

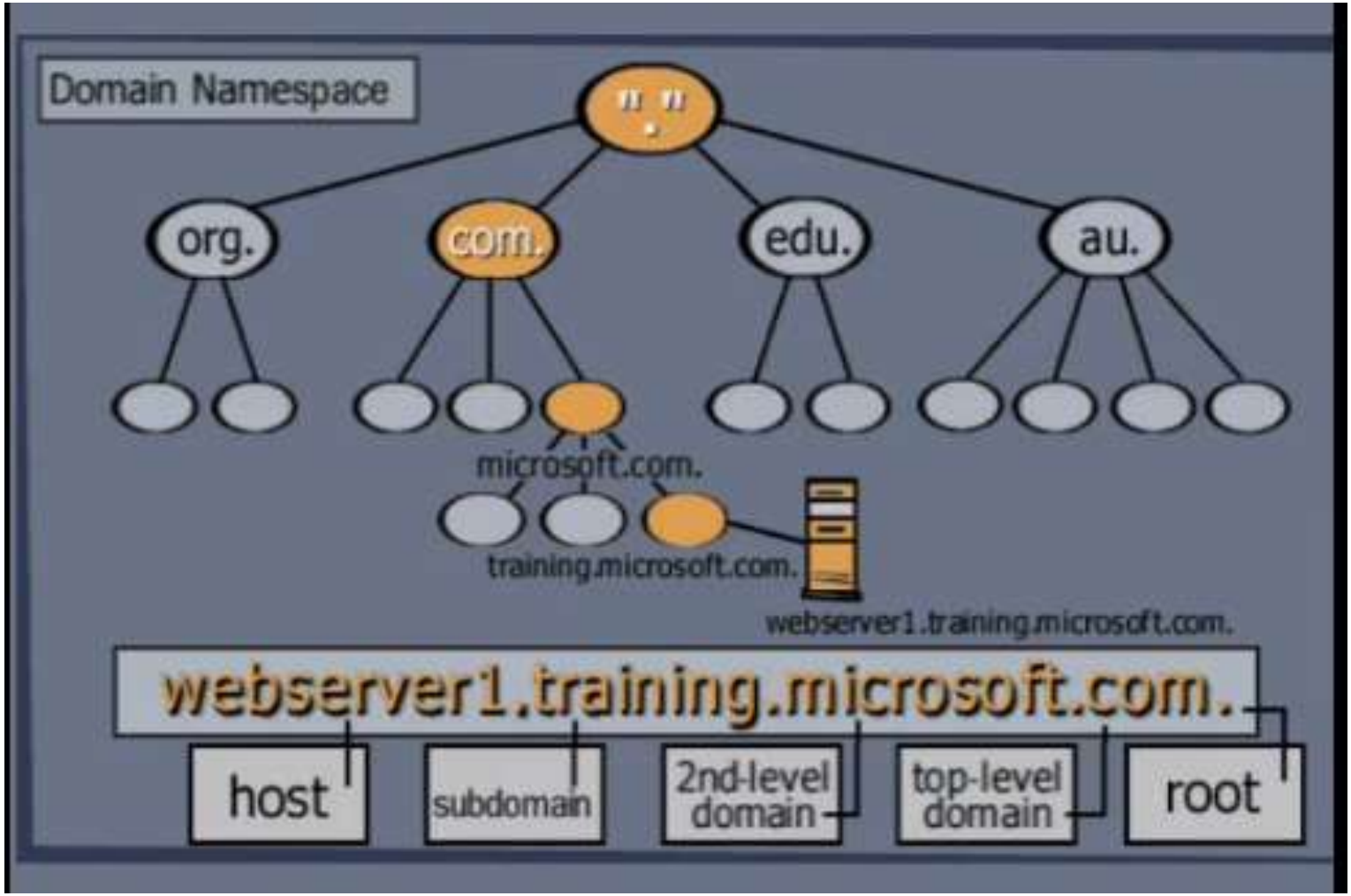
Consistency
server names can
remain constant

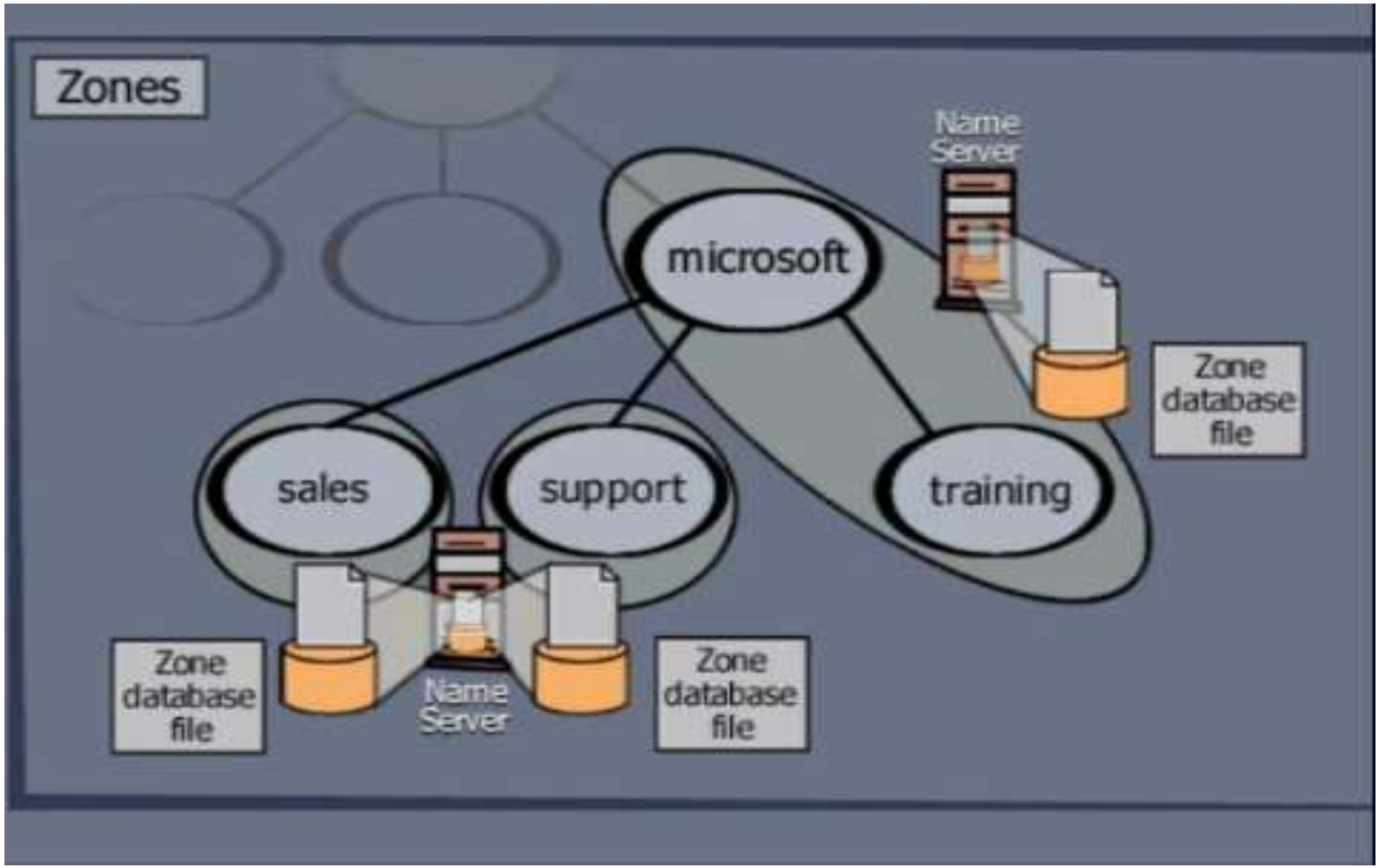
- Introduction to DNS
- Key Components of DNS
  - Domain Namespace
  - Zones
  - Name Servers
- Name Resolution Process

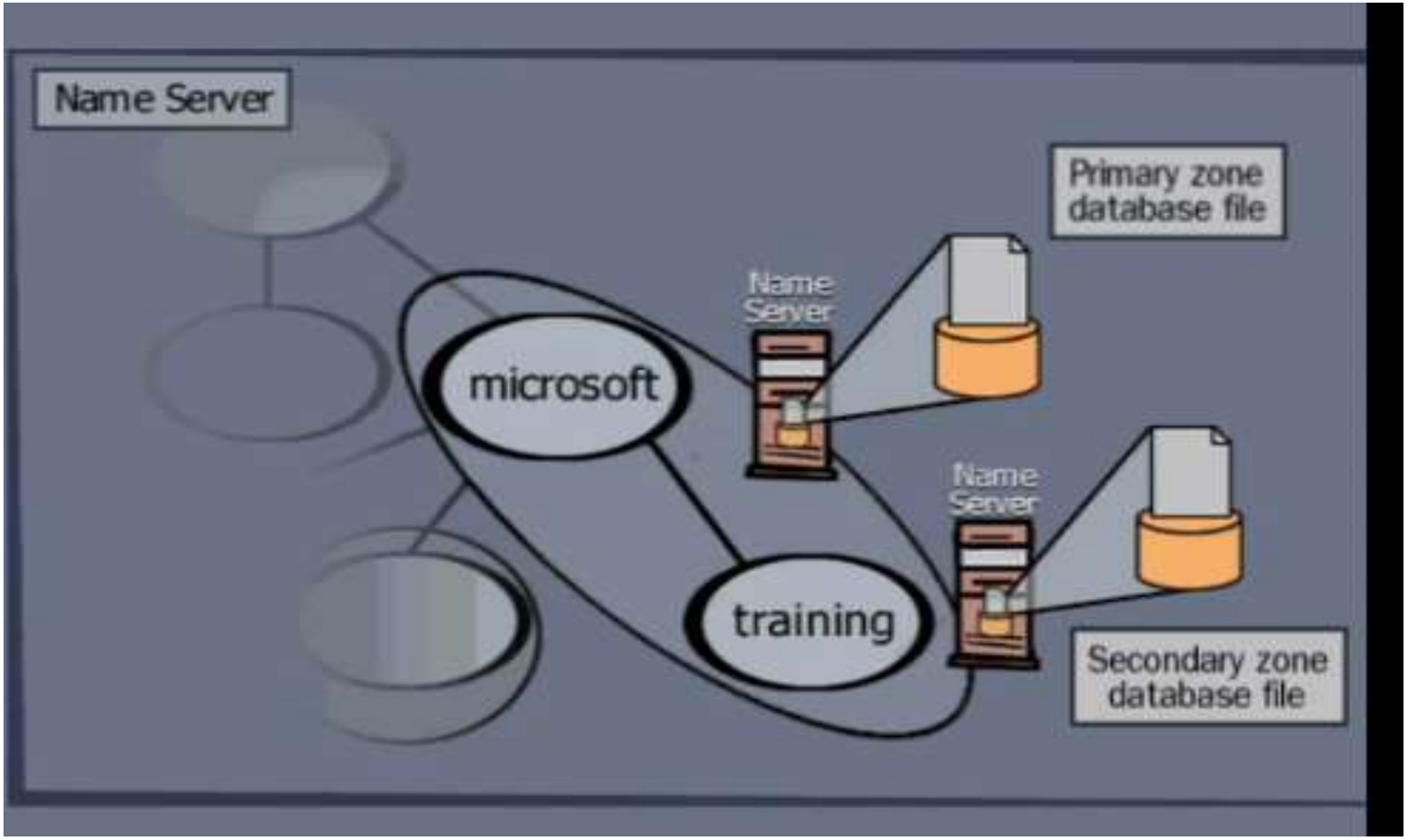

Domain Namespace
" "
org.
com.
edu.
au.
microsoft.com.


Domain Namespace
microsoft.com.
sales.microsoft.com.
support.microsoft.com.
training.microsoft.co

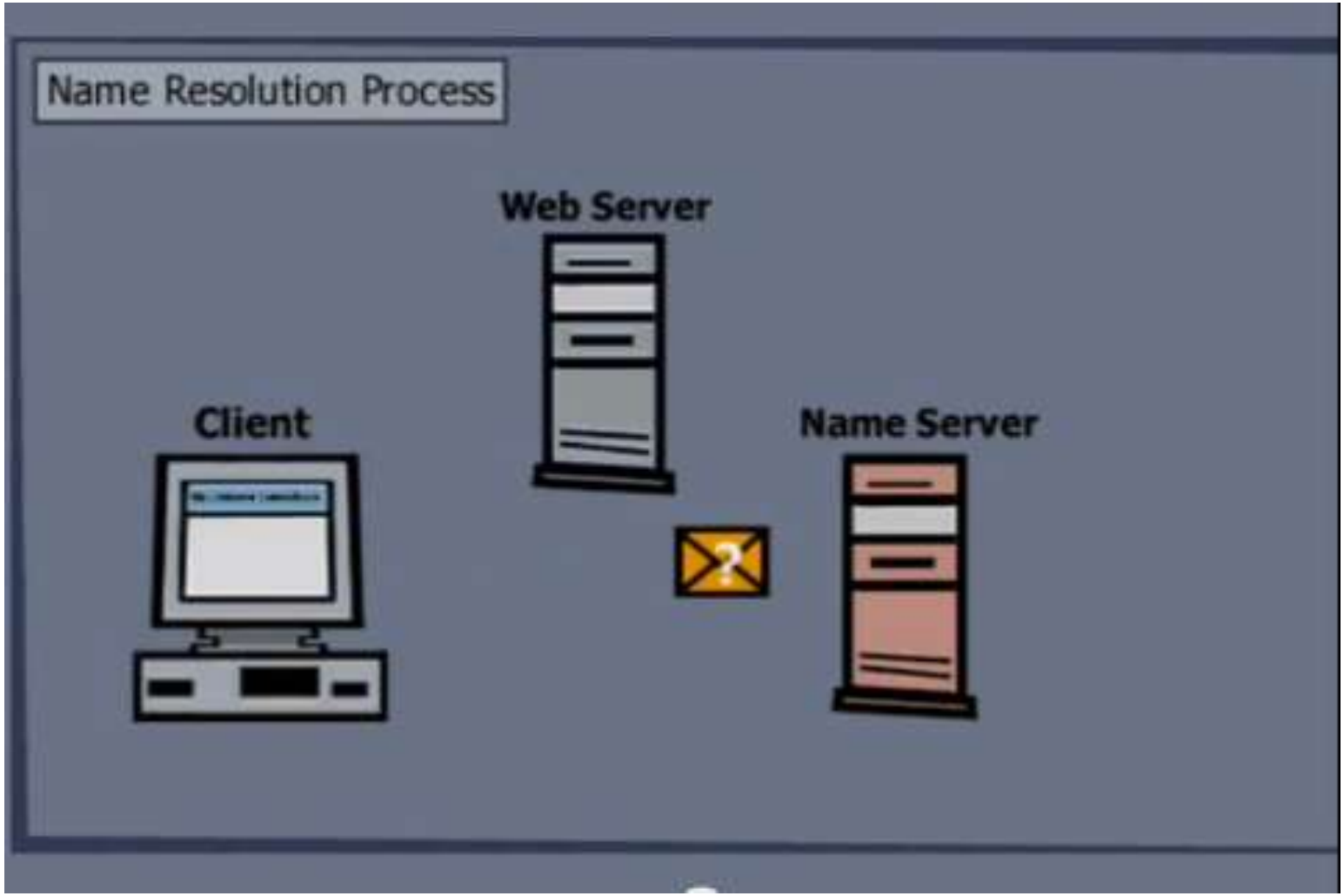

Domain Namespace
microsoft.com.
sales.microsoft.com.
support.microsoft.com.
training.microsoft.co
webserver1.training.microsoft.co
computer1.training.microsoft.com.

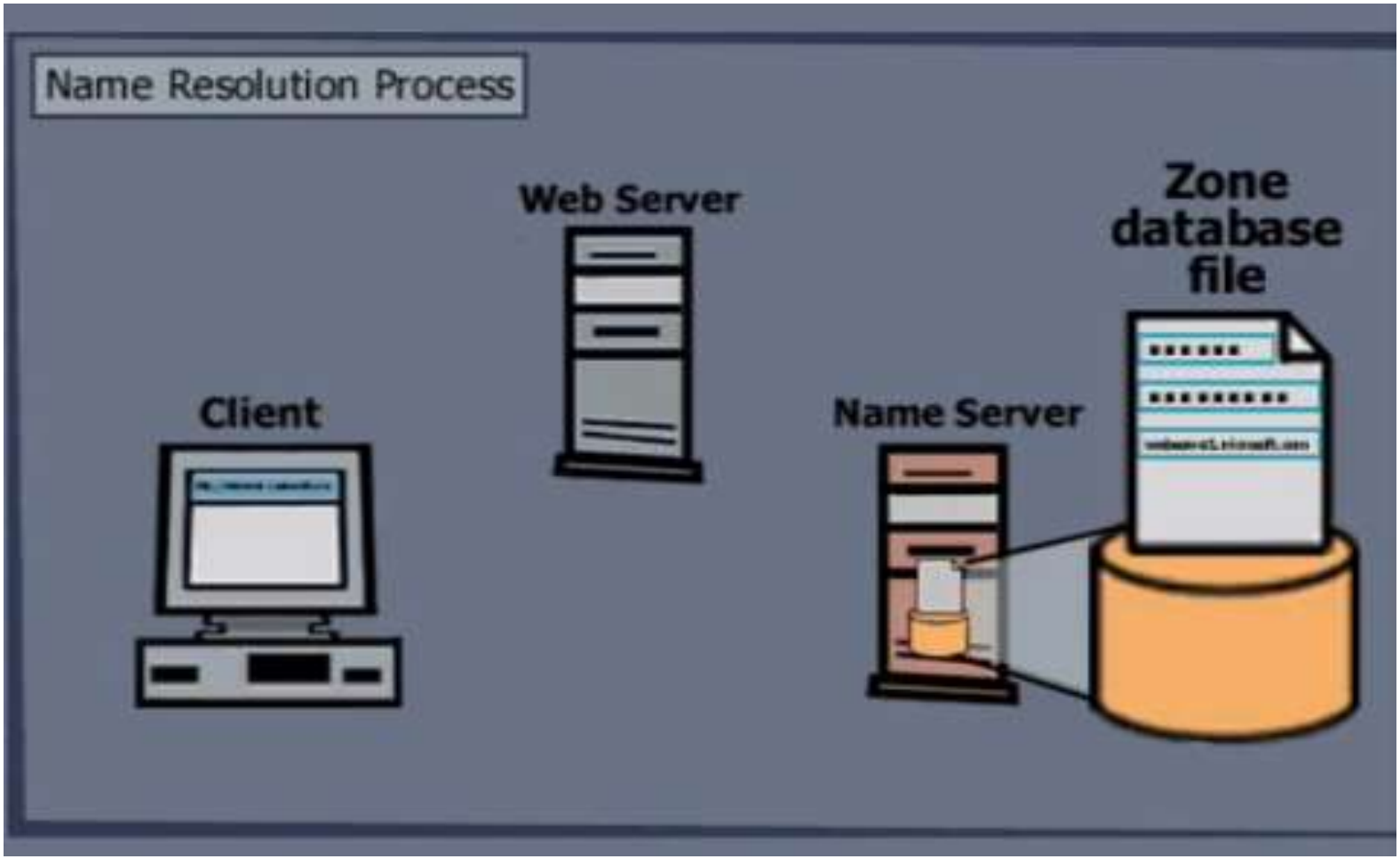

Domain Namespace
" "
org.
com.
edu.
au.
microsoft.com.
training.microsoft.com.
webserver1.training.microsoft.com.
webserver1.training.microsoft.com.
host
subdomain
2nd-level domain
top-level domain
root

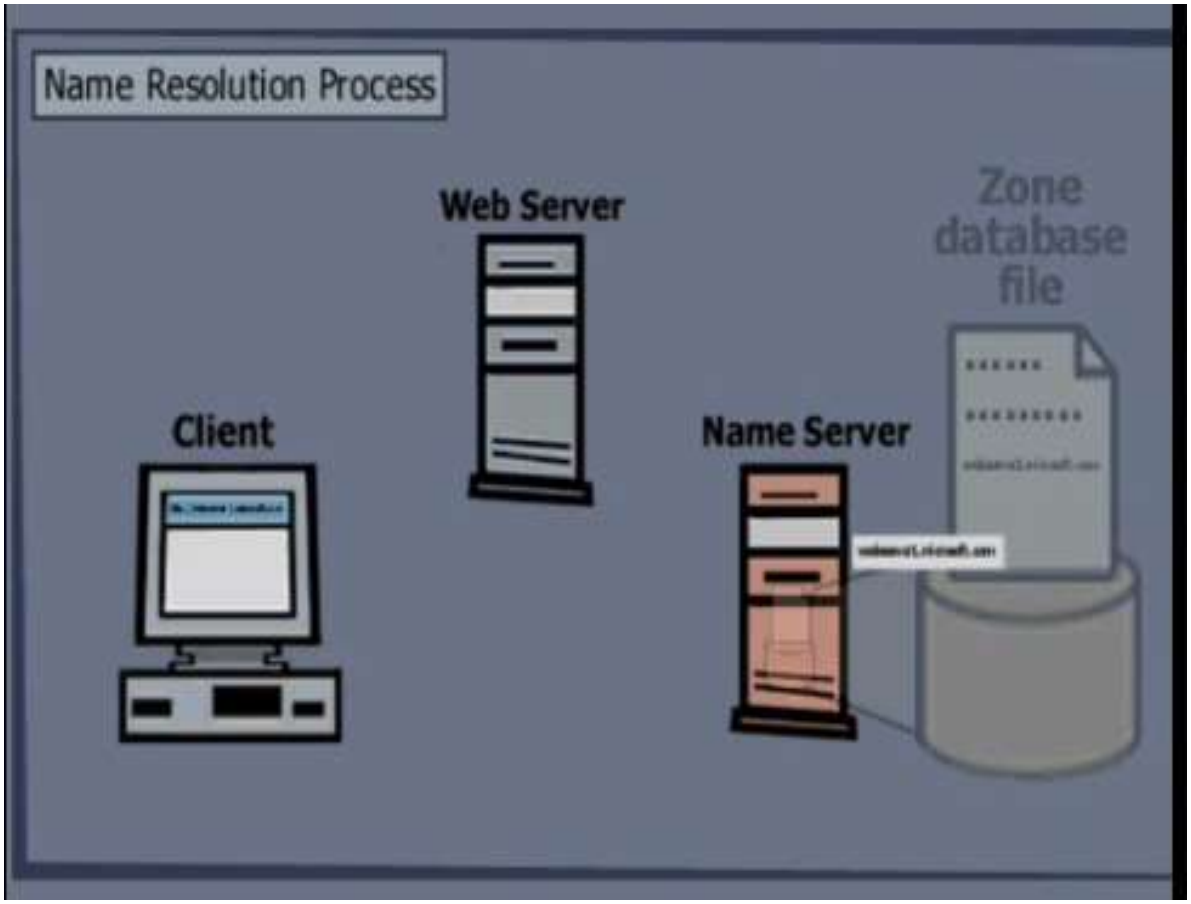

Zones
Name Server
microsoft
Zone database file
sales
support
training
Zone database file
Name Server
Zone database file

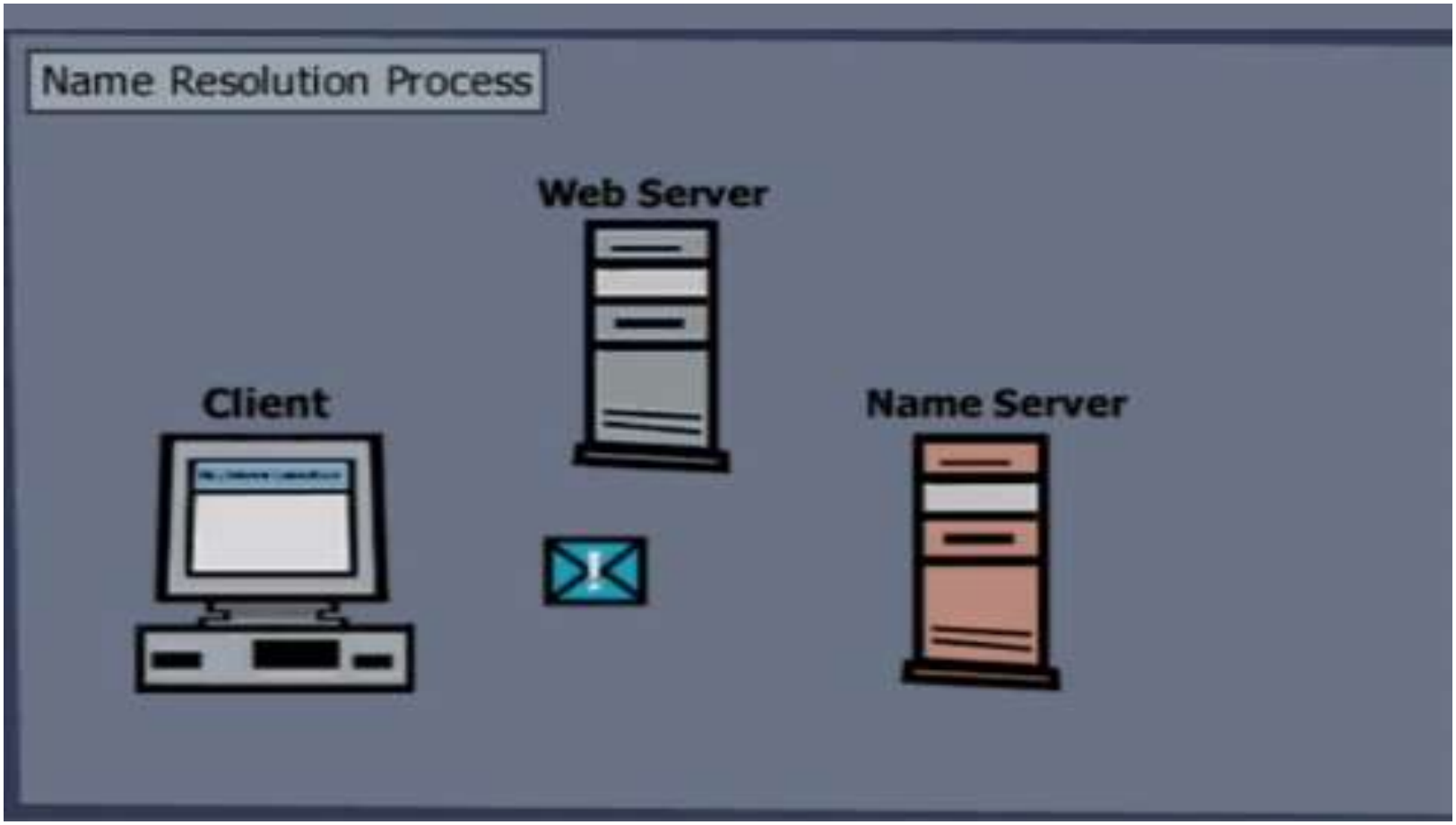

Name Server
Primary zone database file
Name Server
microsoft
Name Server
training
Secondary zone database file

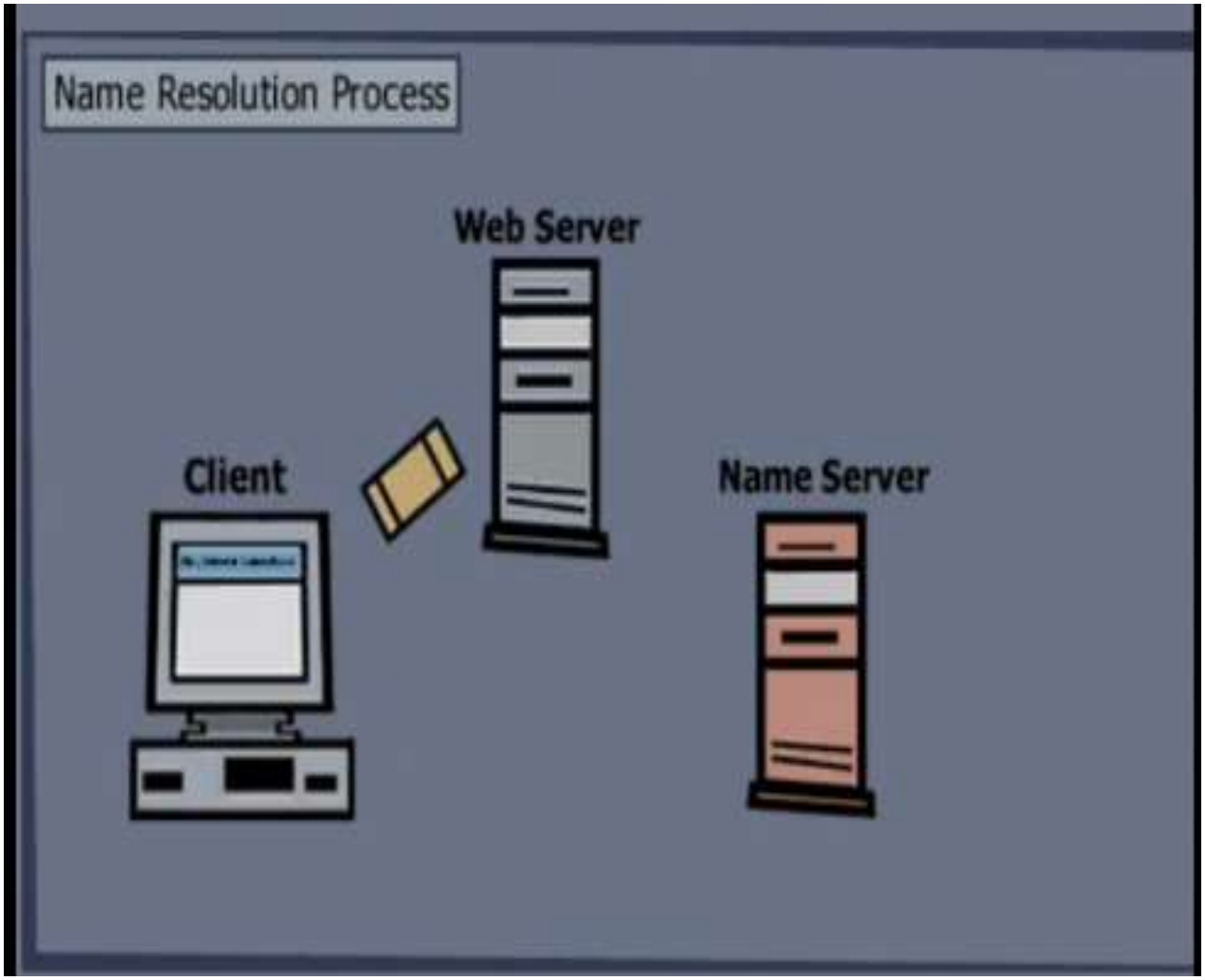
- Introduction to DNS
- Key Components of DNS
- Name Resolution Process

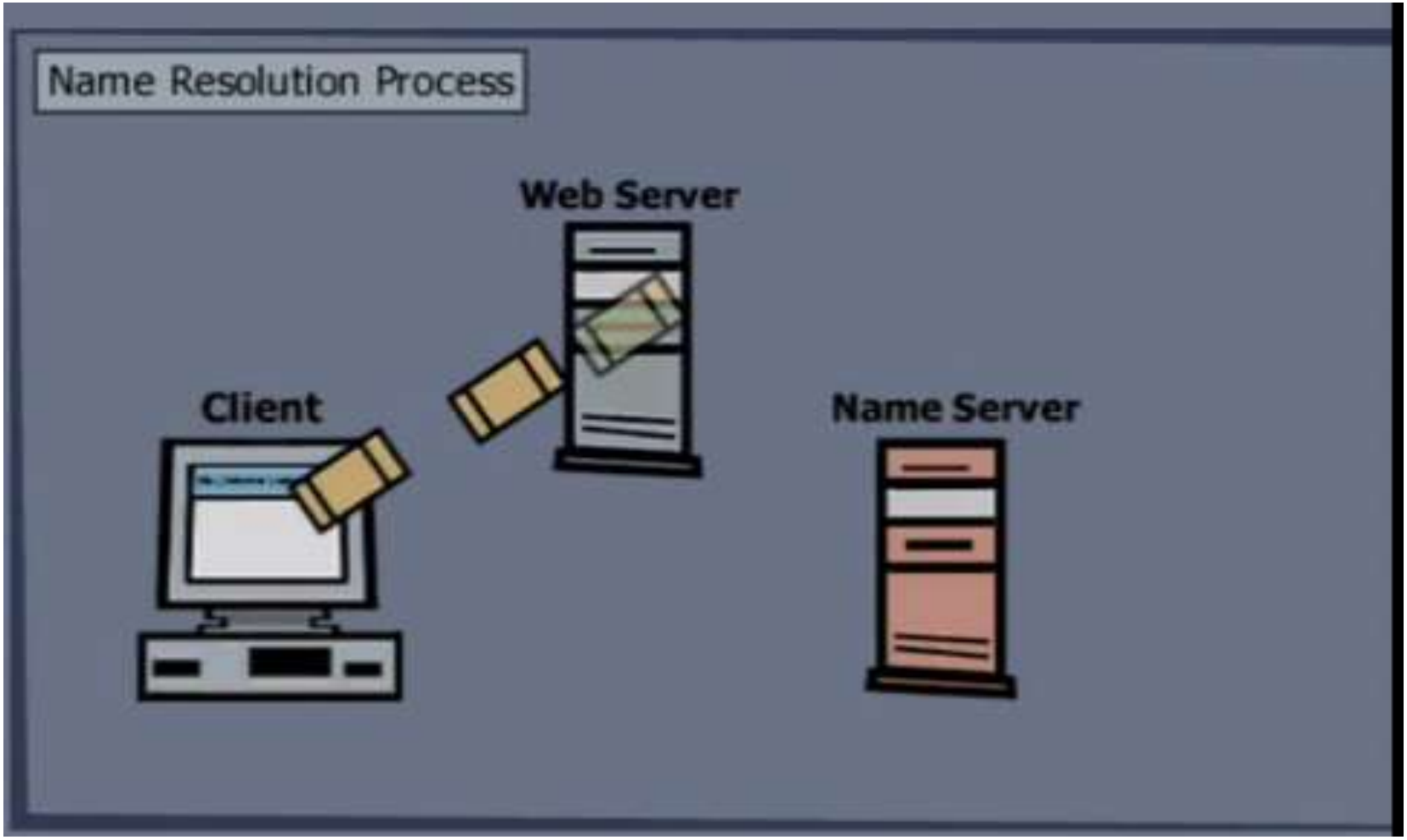

Name Resolution Process
Web Server
Client
Name Server


Name Resolution Process
Web Server
Zone database file
Client
Name Server


Name Resolution Process
Web Server
Zone database file
Client
Name Server

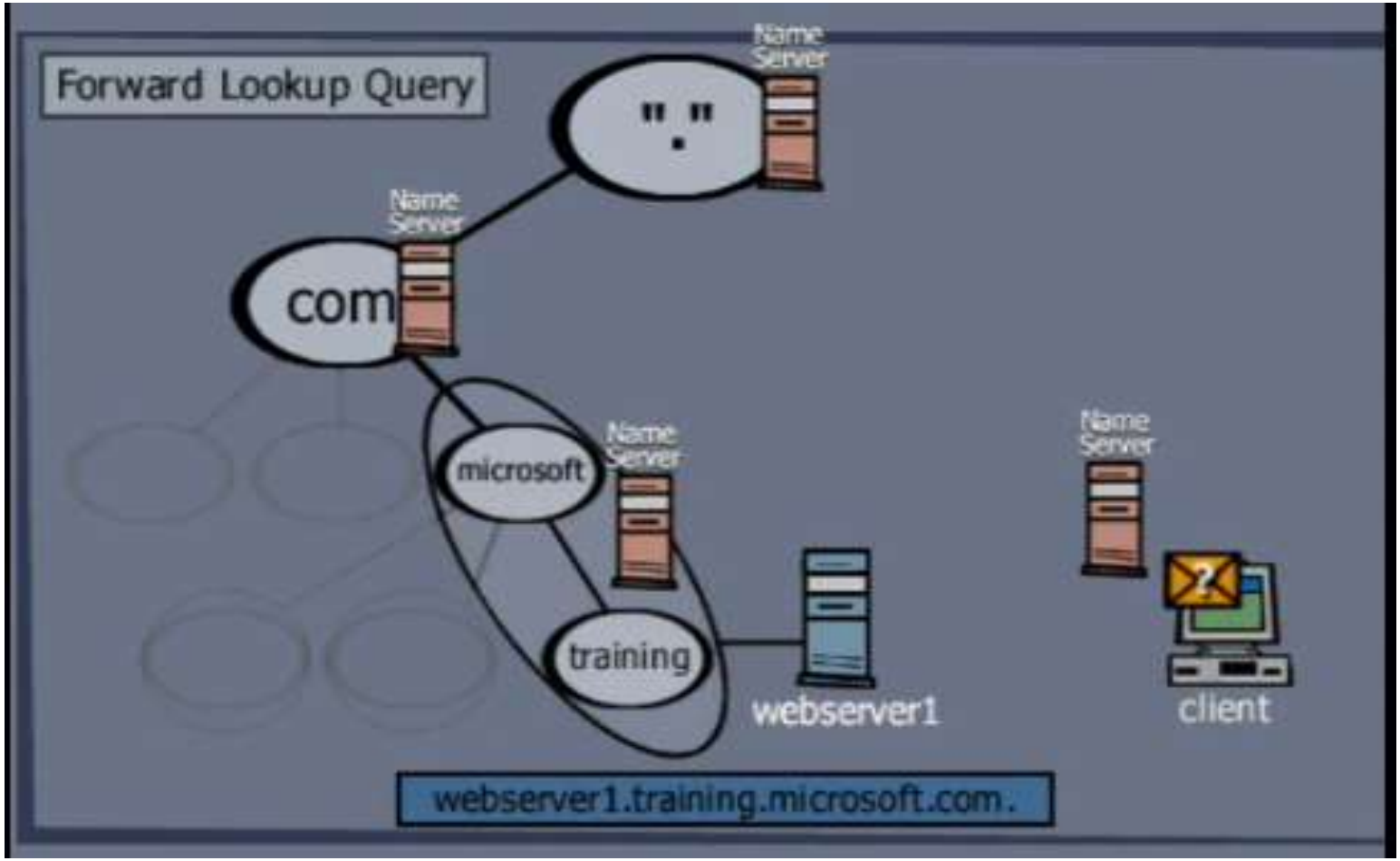

Name Resolution Process
Web Server
Client
Name Server

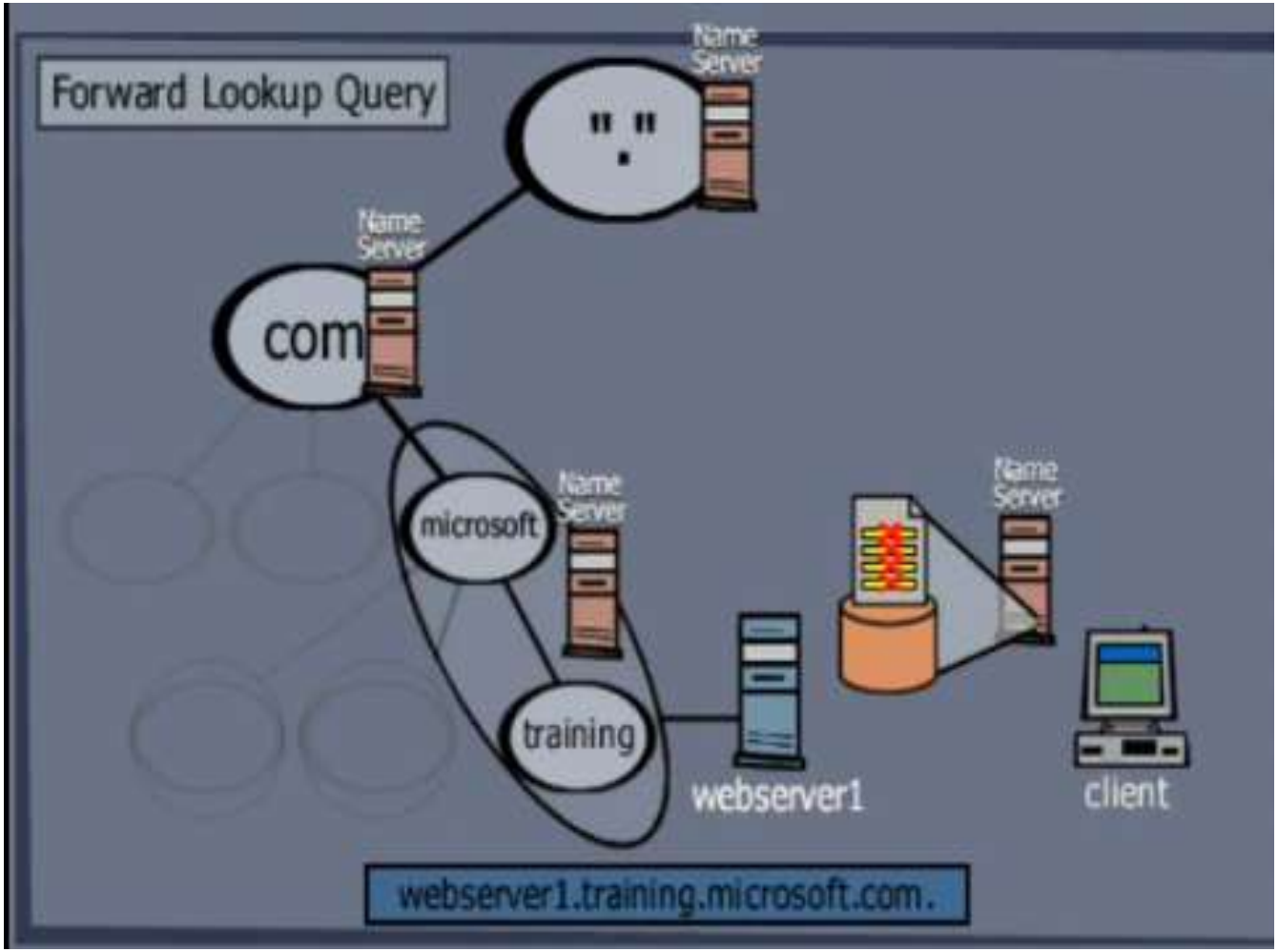

Name Resolution Process
Web Server
Client
Name Server

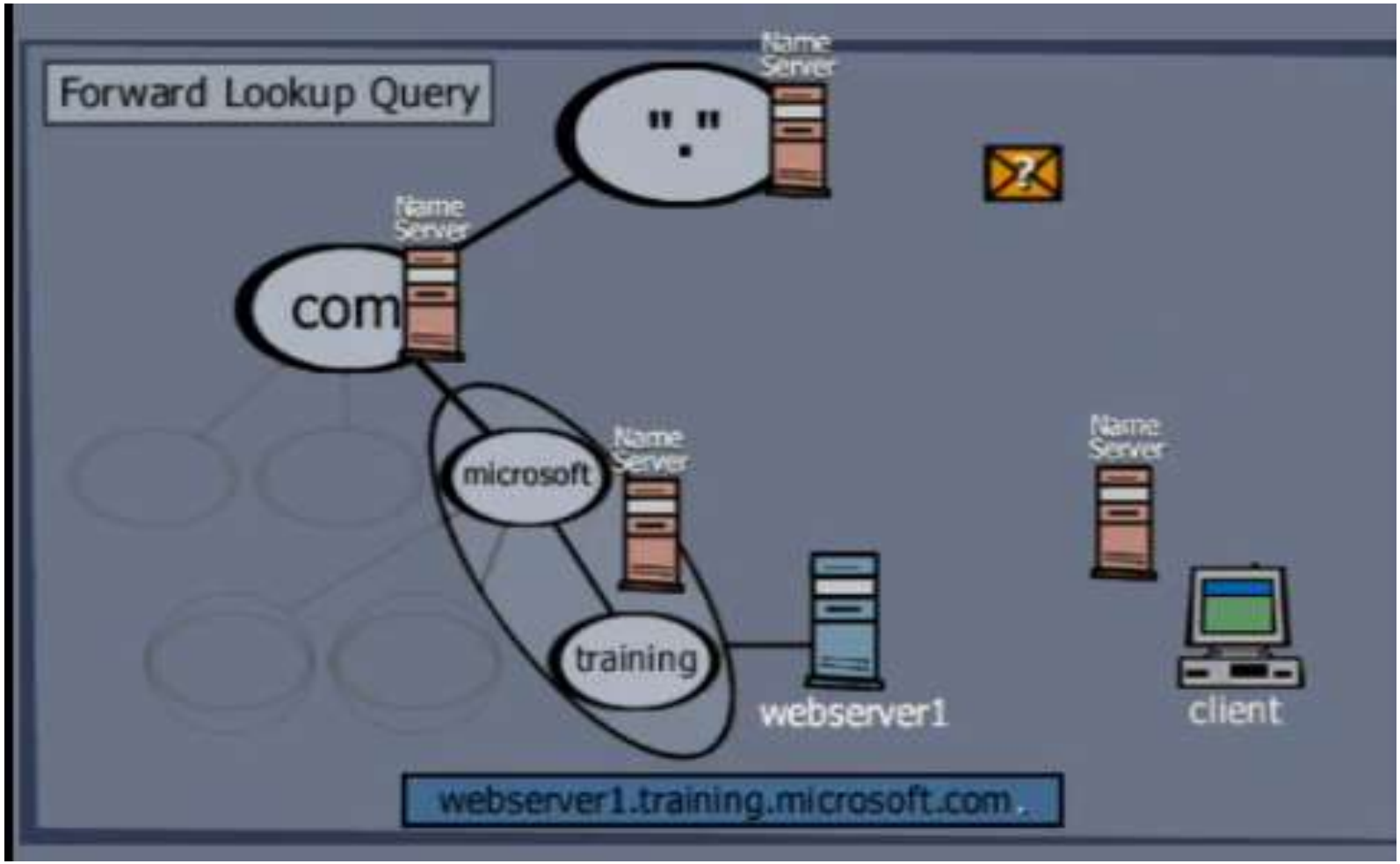

Name Resolution Process
Web Server
Client
Name Server

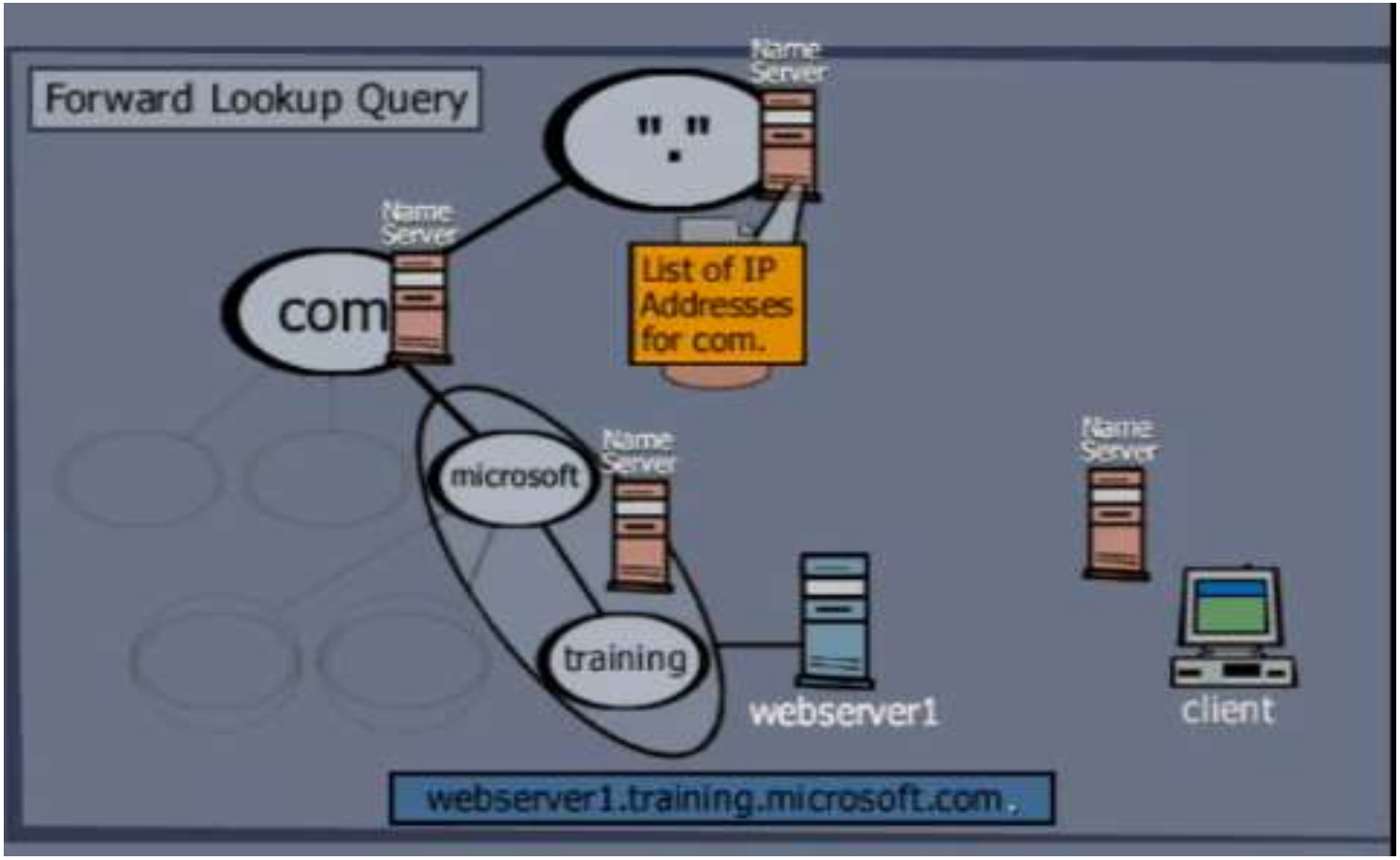
- Name Resolution Process

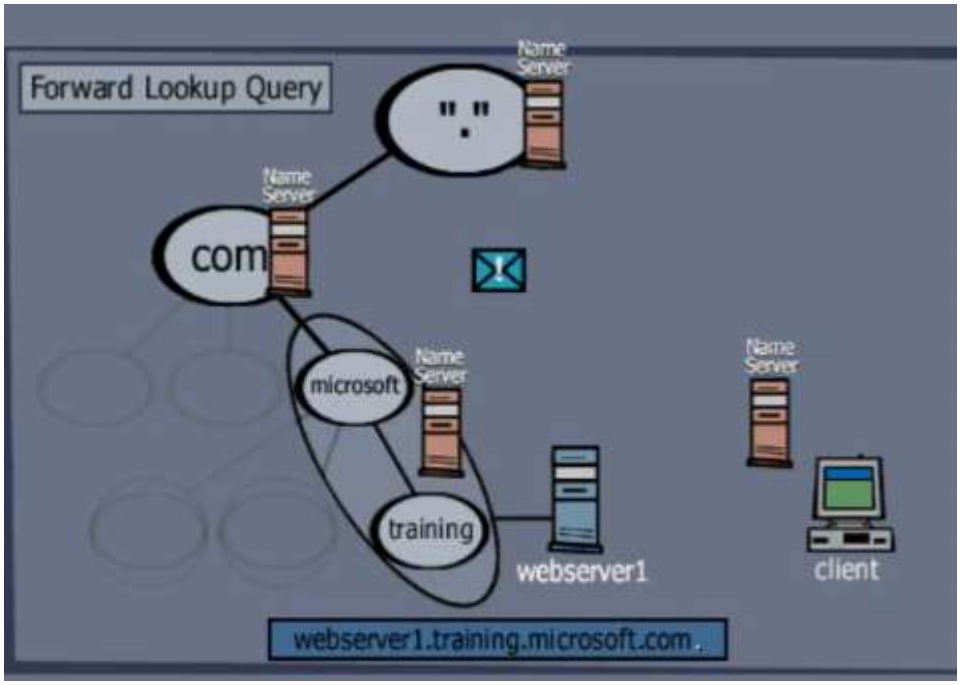
Forward lookup query is a request to map a name to an IP address.

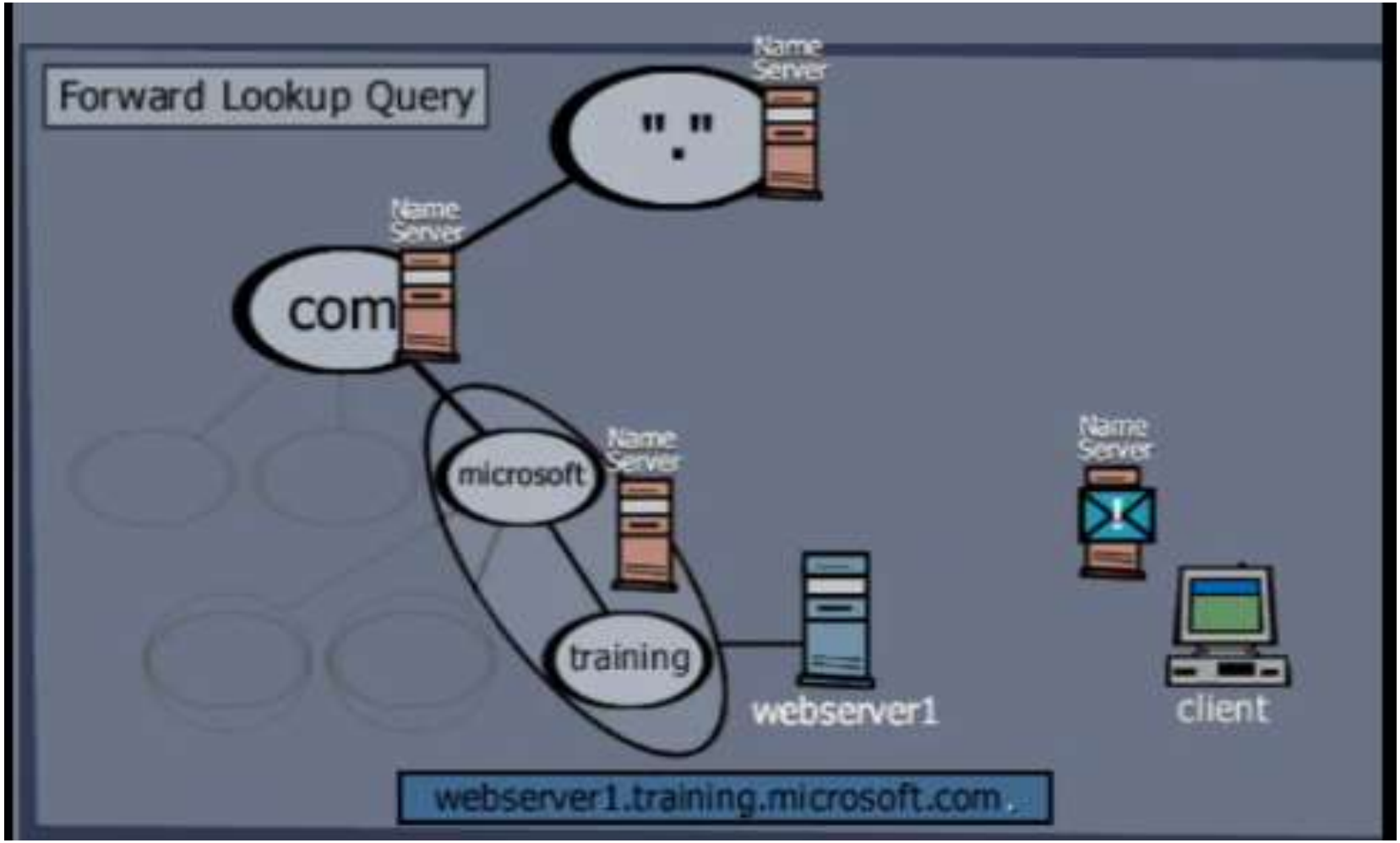

Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

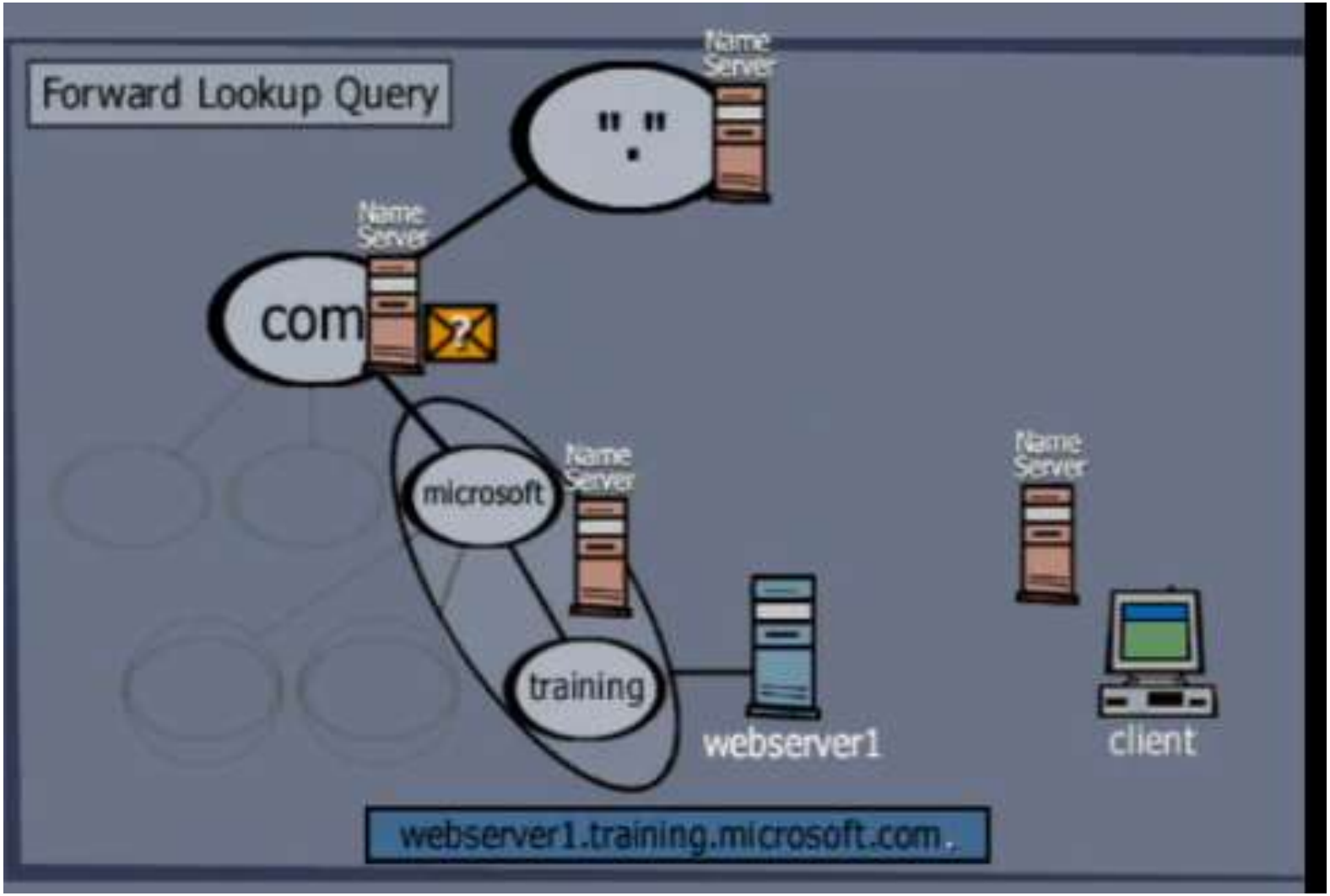

Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

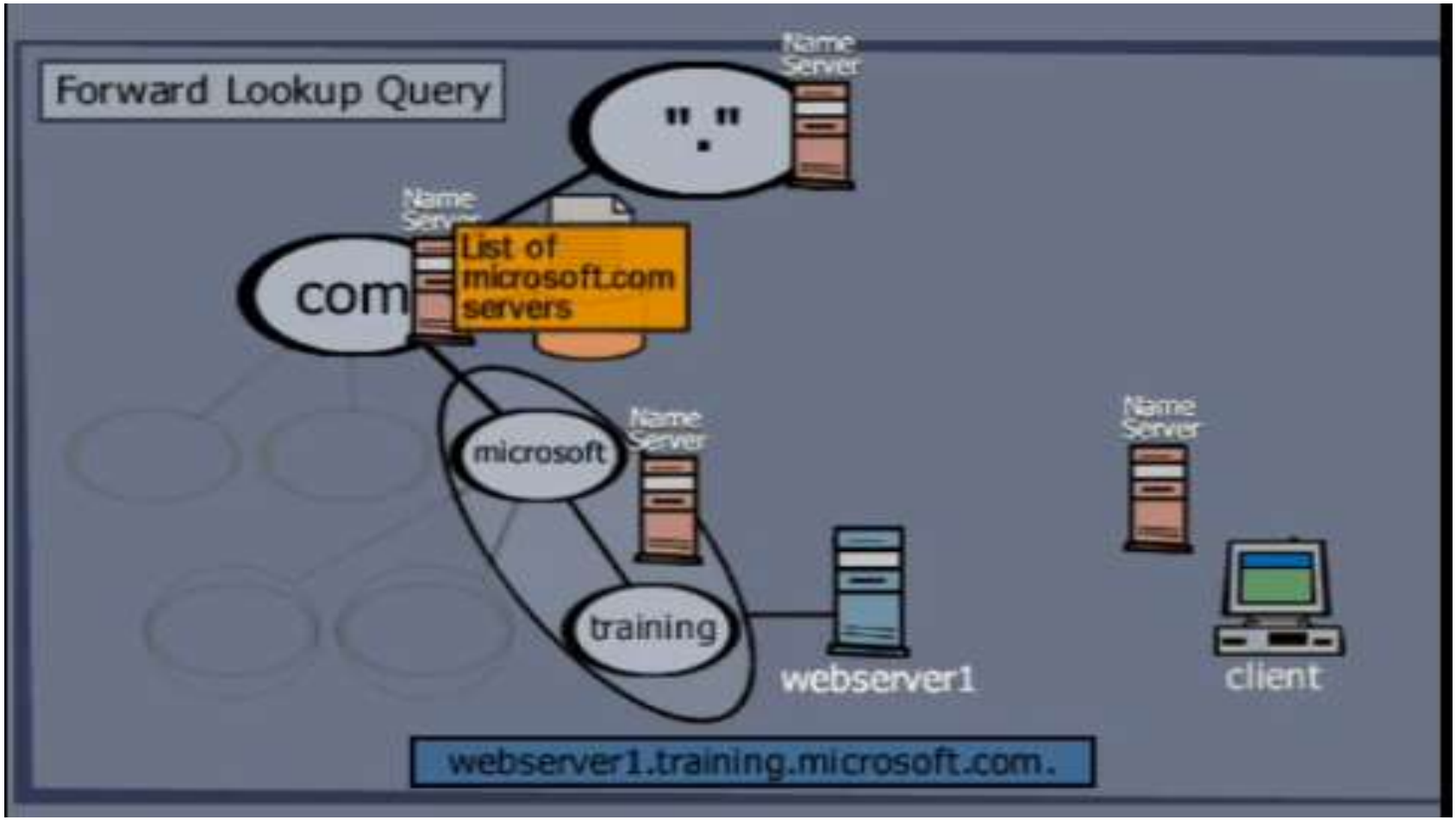

Forward Lookup Query
Name Server
" "
.
?
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

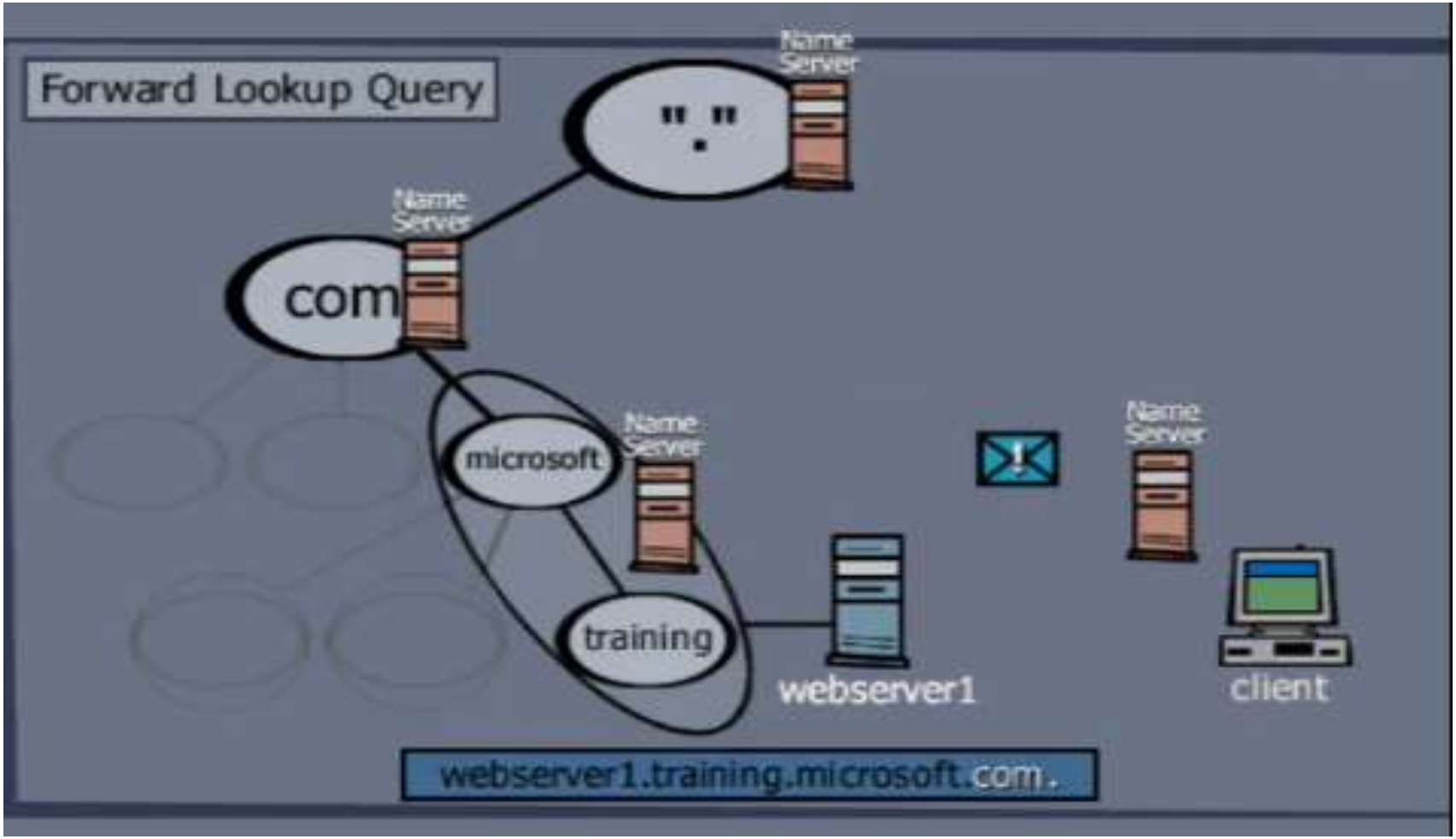

Forward Lookup Query
Name Server
" . "
Name Server
com
List of IP Addresses for com.
microsoft
Name Server
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

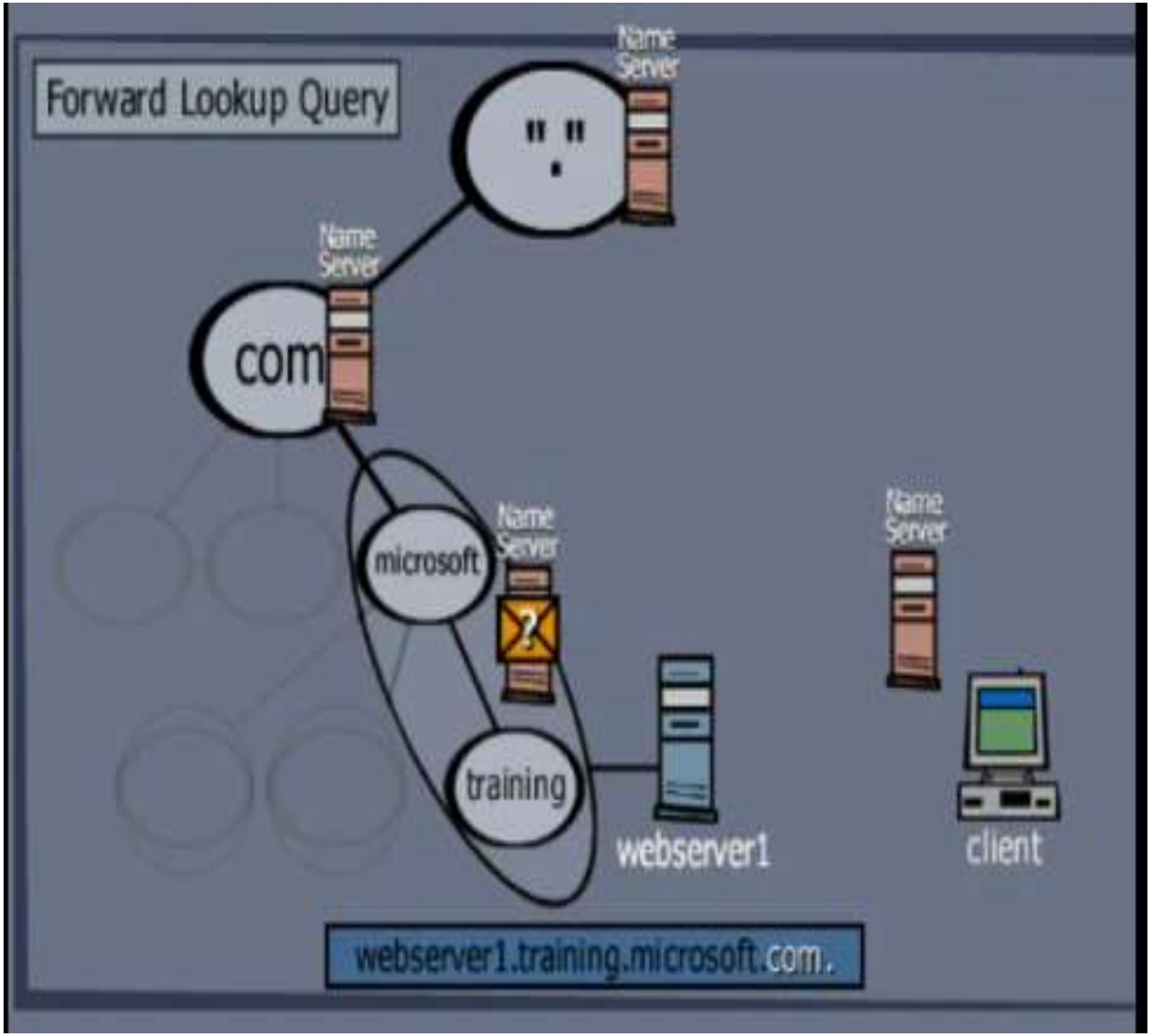

Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

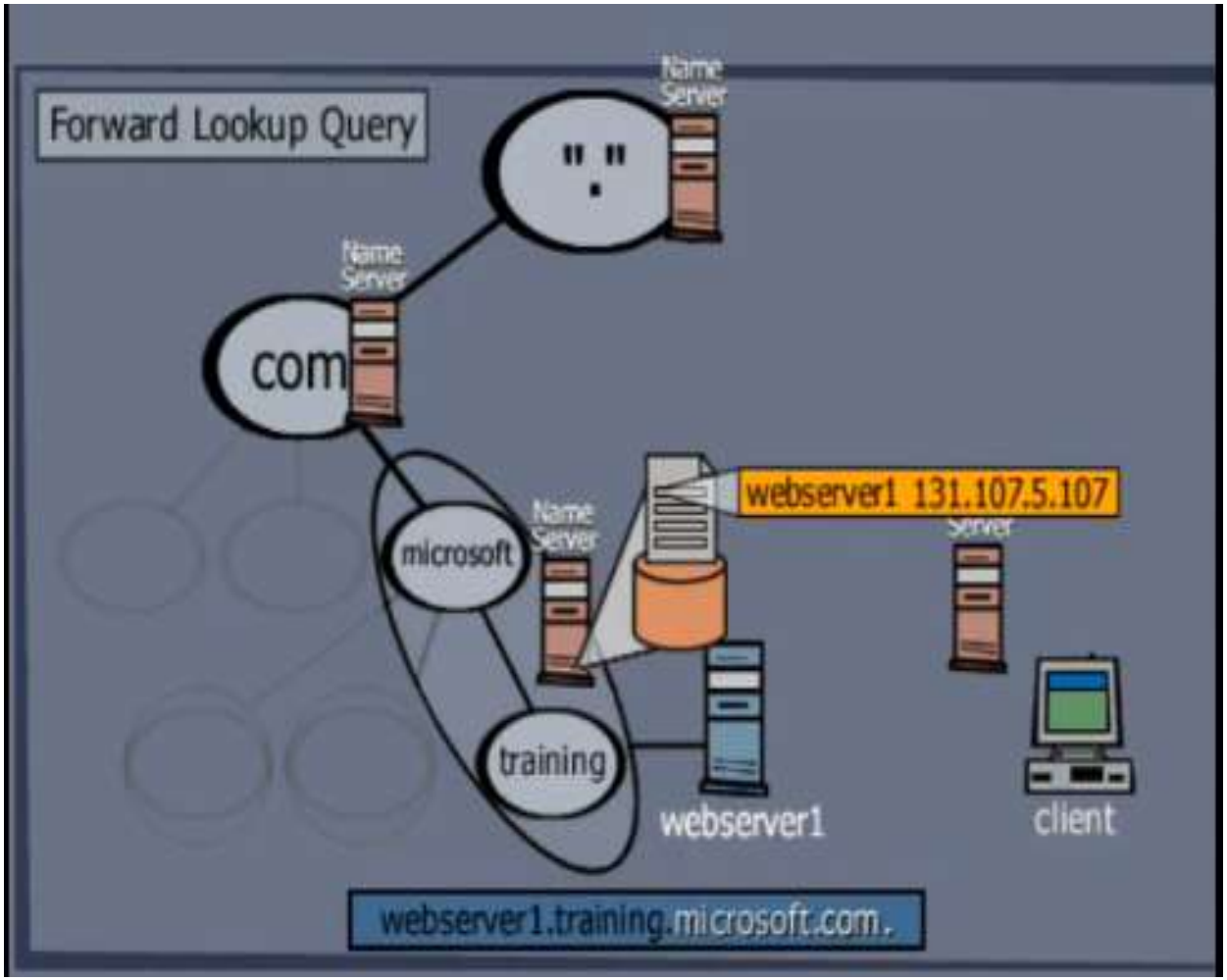

Forward Lookup Query
Name Server
" "
.
Name Server
com
Name Server
microsoft
training
webserver1
Name Server
client
webserver1.training.microsoft.com.

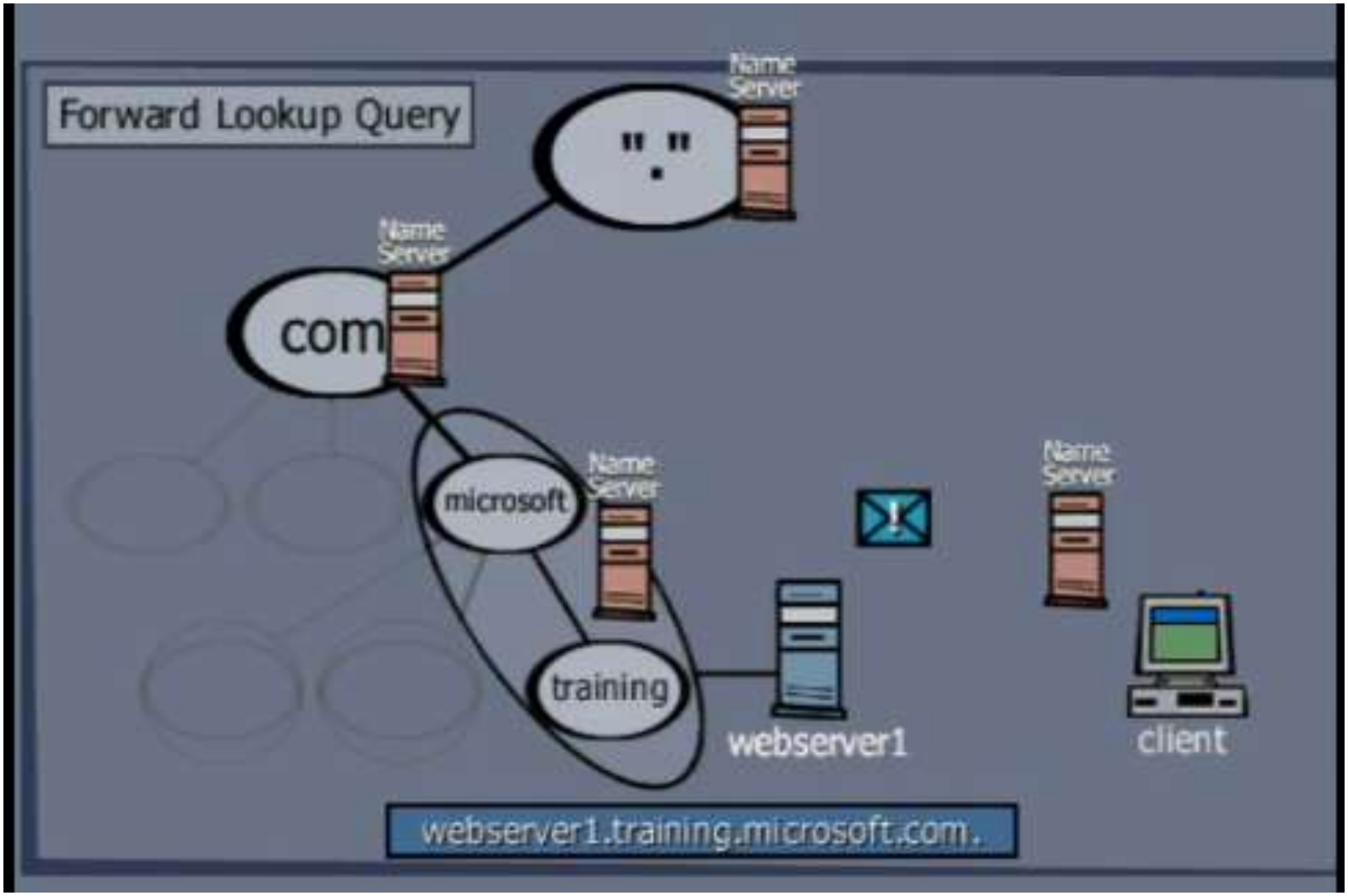

Forward Lookup Query
Name Server
" "
Name Server
com
?
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

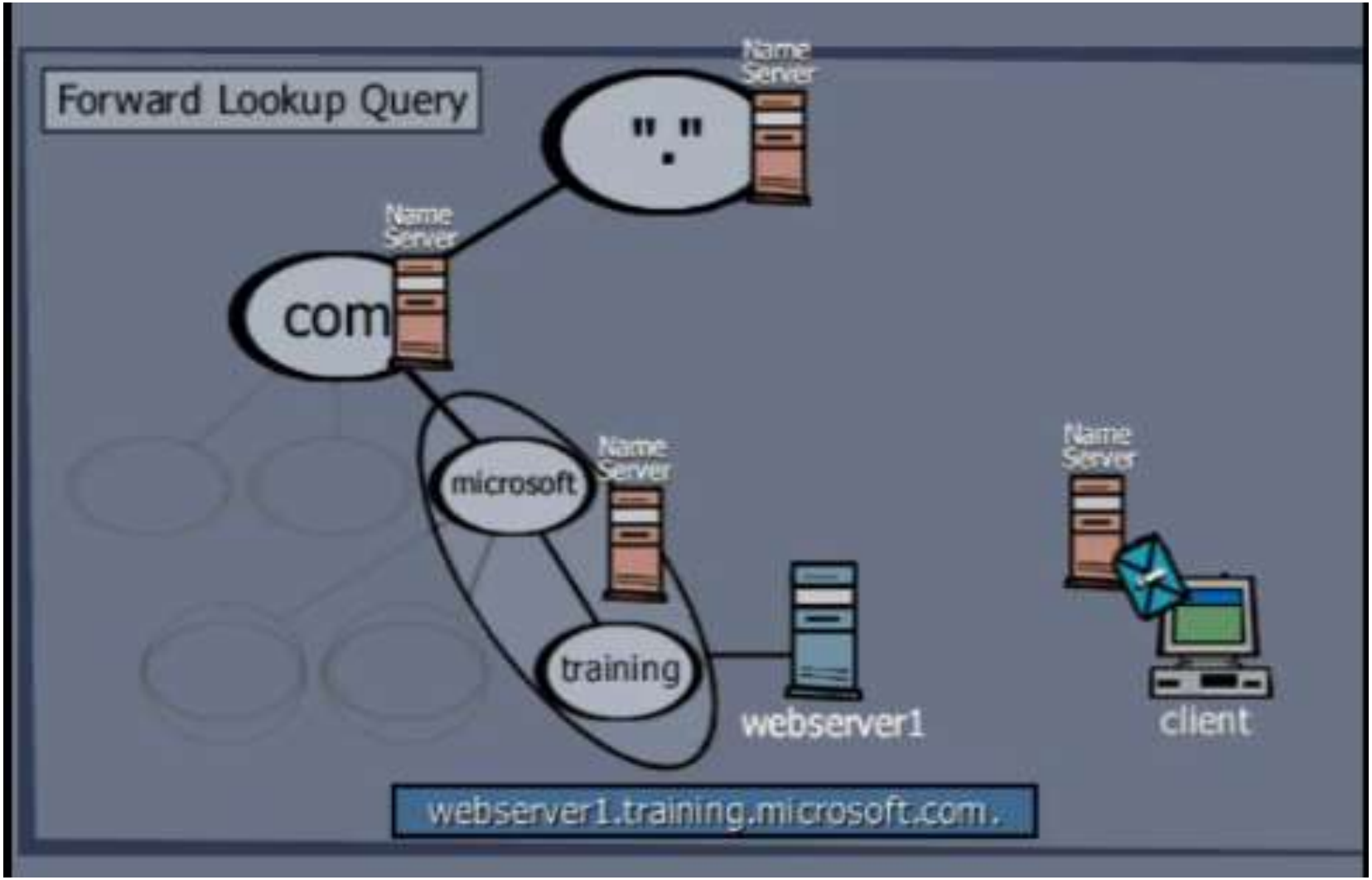

Forward Lookup Query
Name Server
" . "
Name Server
List of microsoft.com servers
com
microsoft
Name Server
training
webserver1
Name Server
client
webserver1.training.microsoft.com.

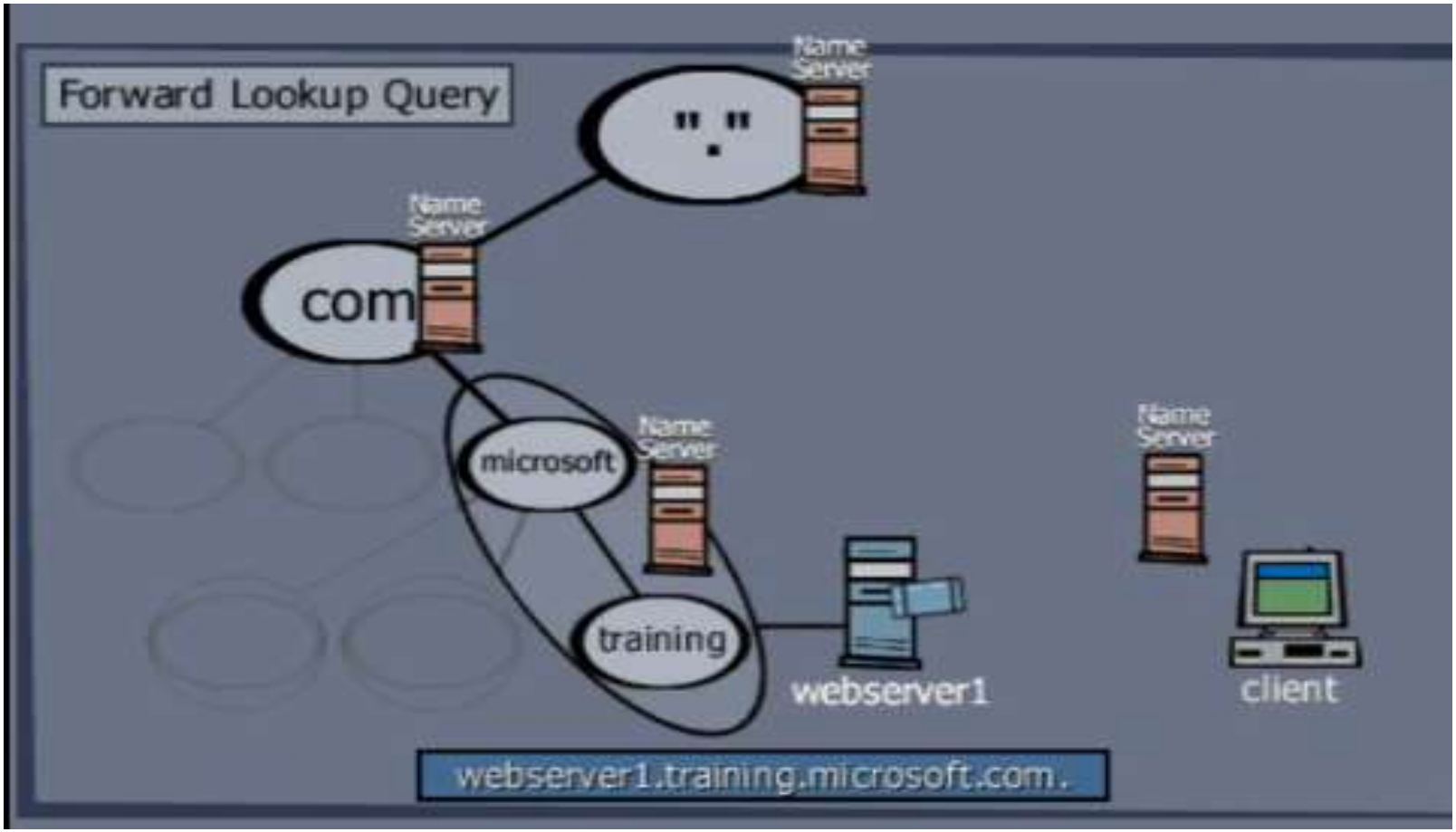

Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.

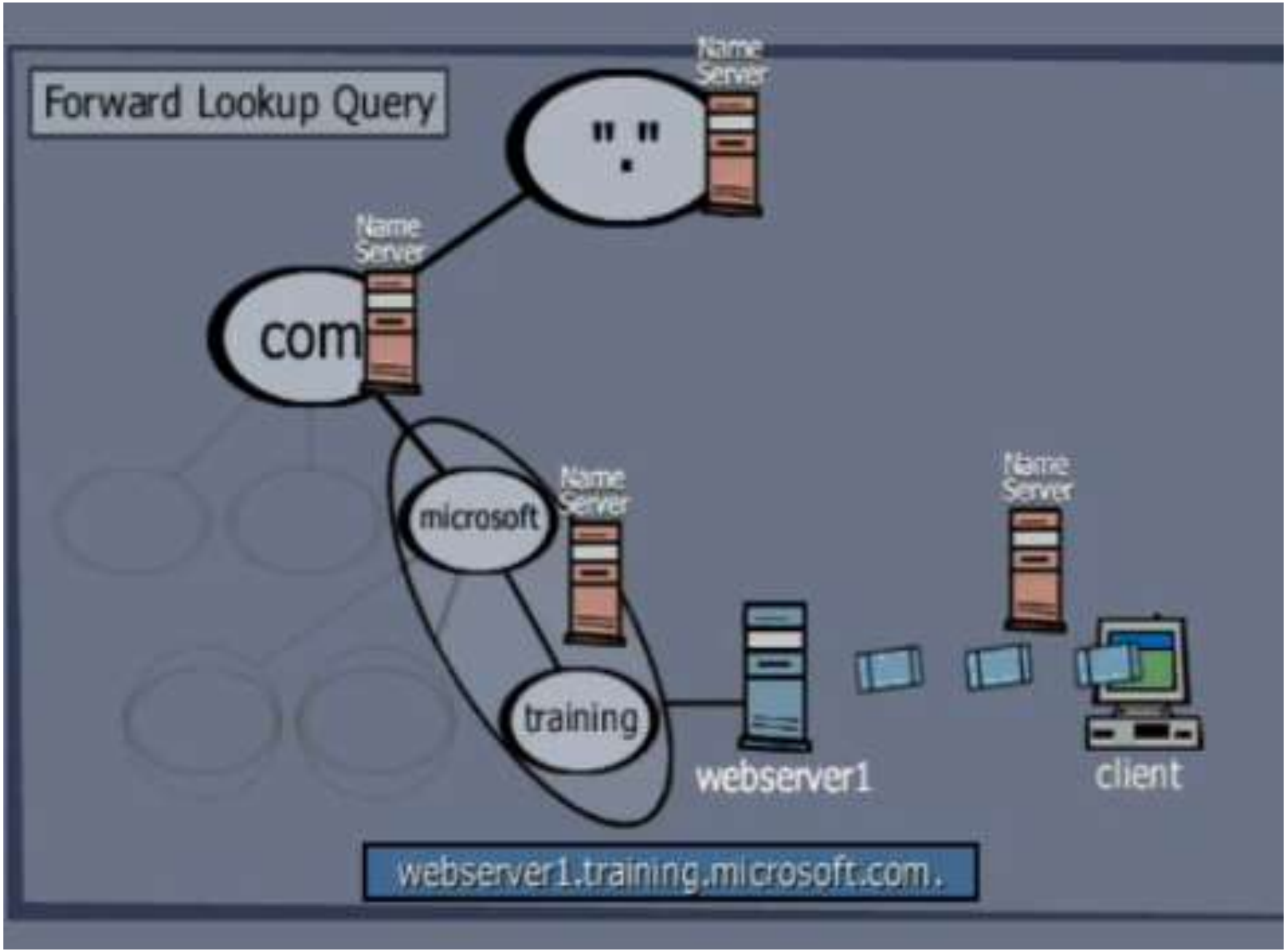

Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
?
training
webserver1
Name Server
client
webserver1.training.microsoft.com.

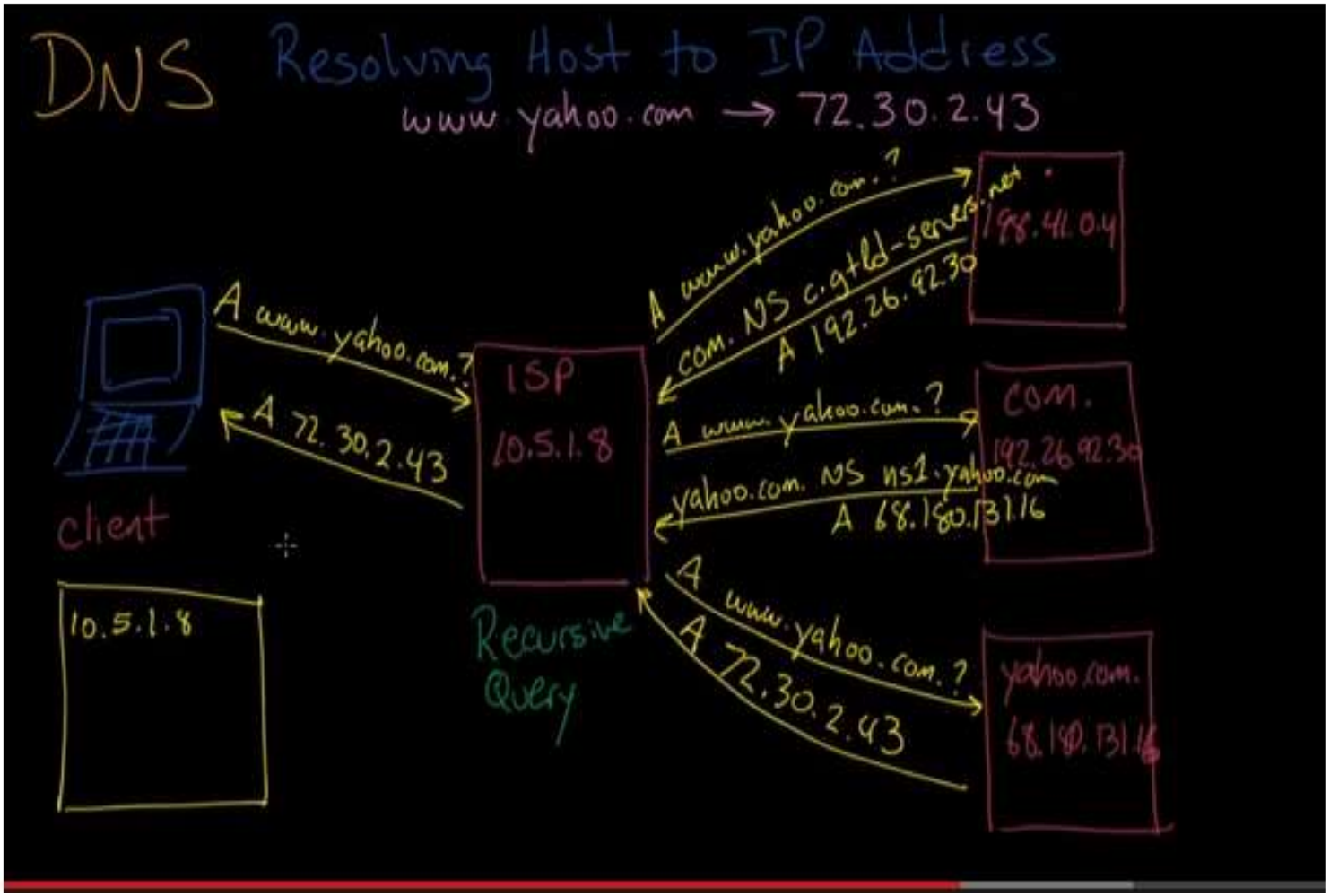

Forward Lookup Query
Name Server
" "
Name Server
com
microsoft
Name Server
webserver1 131.107.5.107
training
webserver1
client
webserver1.training.microsoft.com.

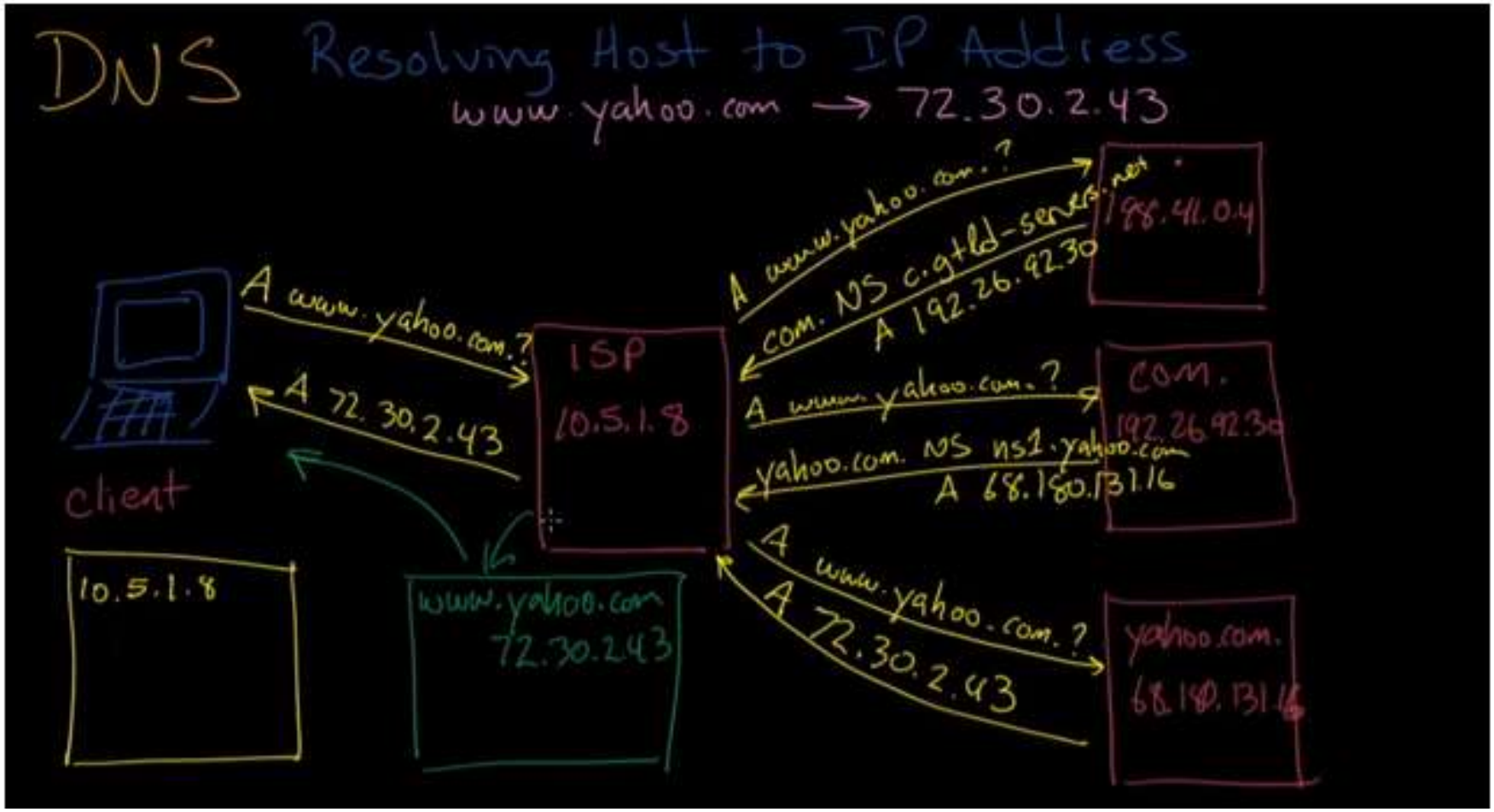

Forward Lookup Query
Name Server
" "
.
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.


Forward Lookup Query
Name Server
" "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
Name Server
client
webserver1.training.microsoft.com.


Forward Lookup Query
Name Server
" . "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.


Forward Lookup Query
Name Server
" . "
Name Server
com
Name Server
microsoft
Name Server
training
webserver1
client
webserver1.training.microsoft.com.


DNS
Resolving Host to IP Address
www.yahoo.com → 72.30.2.43
A www.yahoo.com.?
A 72.30.2.43
client
10.5.1.8
ISP
10.5.1.8
Recursive Query
A www.yahoo.com.?
com. NS c.gtld-servers.net
A 192.26.92.30
198.41.0.4
A www.yahoo.com.?
yahoo.com. NS ns1.yahoo.com
A 68.180.131.16
com.
192.26.92.30
A www.yahoo.com.?
A 72.30.2.43
yahoo.com.
68.180.131.16


DNS
Resolving Host to IP Address
www.yahoo.com → 72.30.2.43
client
A www.yahoo.com.?
A 72.30.2.43
ISP
10.5.1.8
A www.yahoo.com.?
com. NS c.gtld-servers.net
A 192.26.92.30
198.41.0.4
A www.yahoo.com.?
yahoo.com. NS ns1.yahoo.com
A 68.180.131.16
COM.
192.26.92.30
A www.yahoo.com.?
A 72.30.2.43
yahoo.com.
68.180.131.16
10.5.1.8
www.yahoo.com
72.30.2.43

DNS

Domains form a tree.

google.com

www.google.com

www.yahoo.com

aspmx.l.google.com

www.ucsd.edu

www.cs.ucsd.edu

www.bbc.co.uk

DNS

Domains form a tree.

google.com

www.google.com

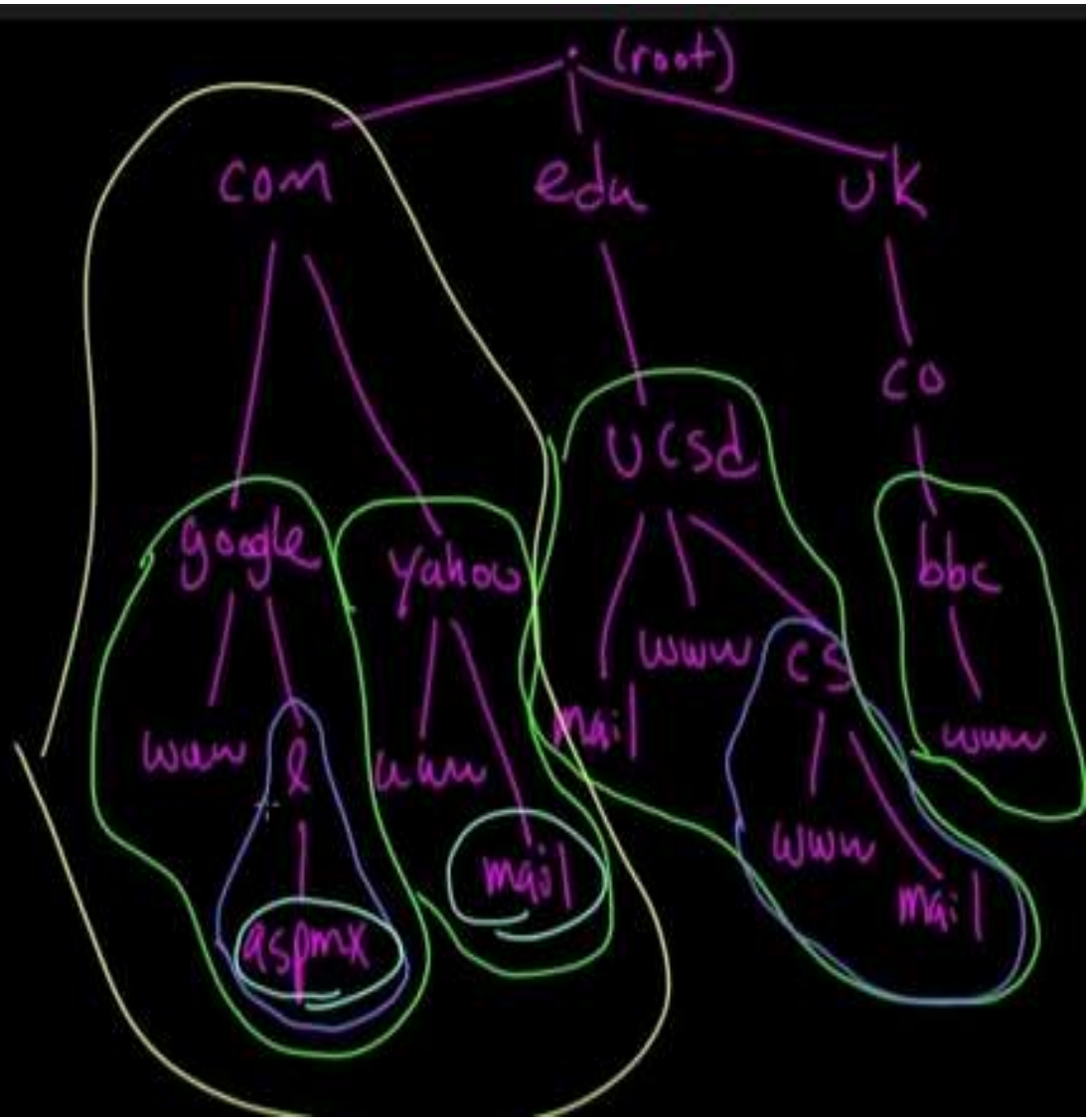
www.yahoo.com

aspmx.l.google.com

www.ucsd.edu

www.cs.ucsd.edu

www.bbc.co.uk



DNS Terminology

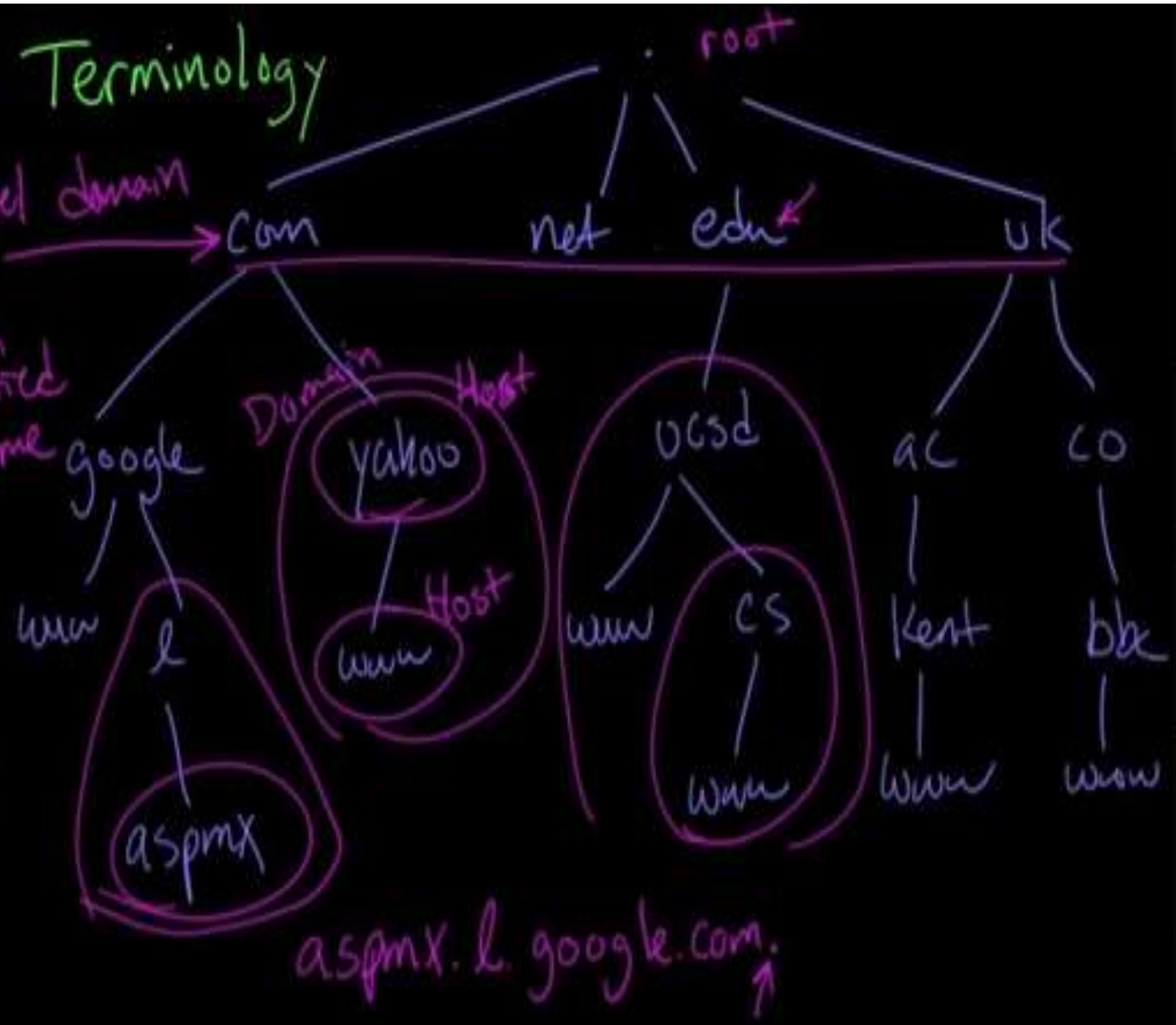
TLD = top level domain

FQDN = Fully qualified domain name

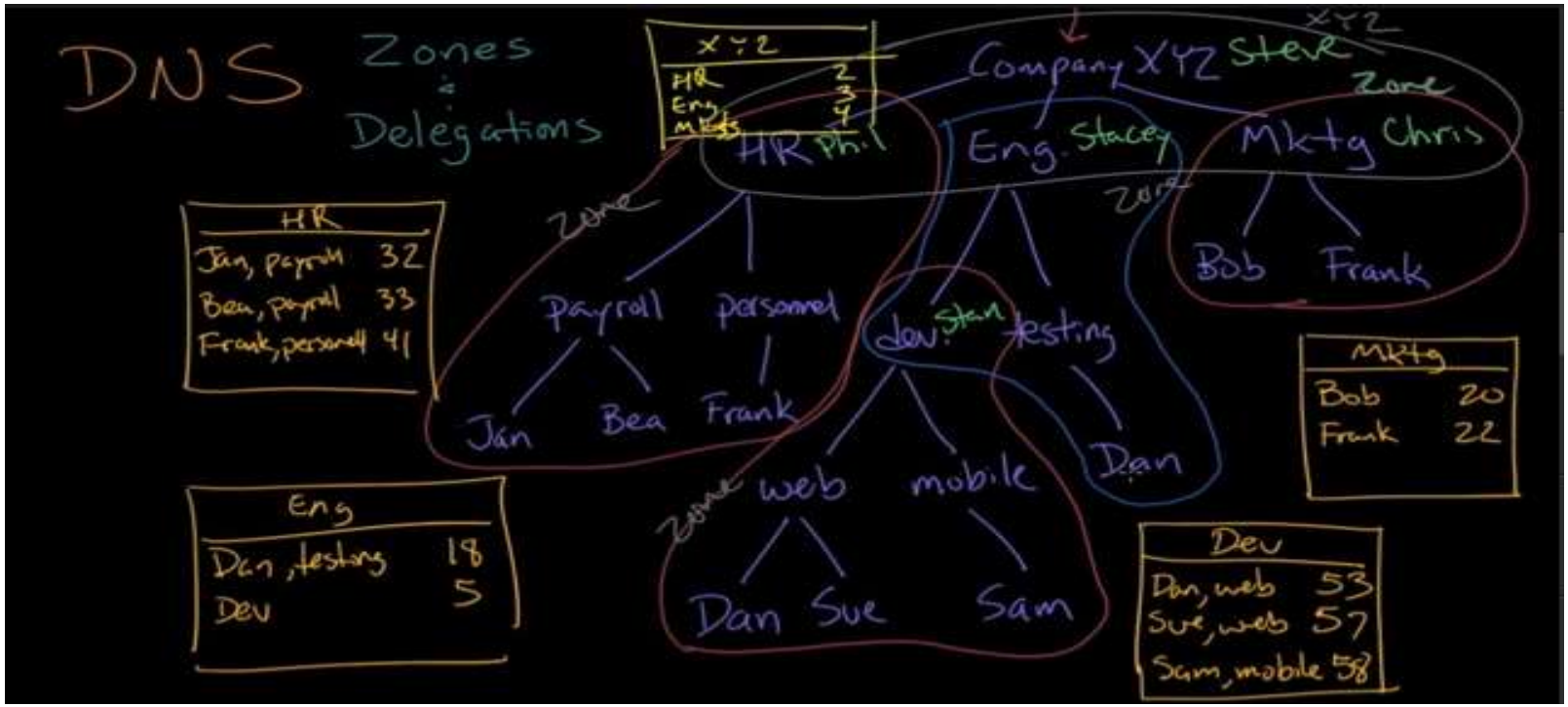
Host

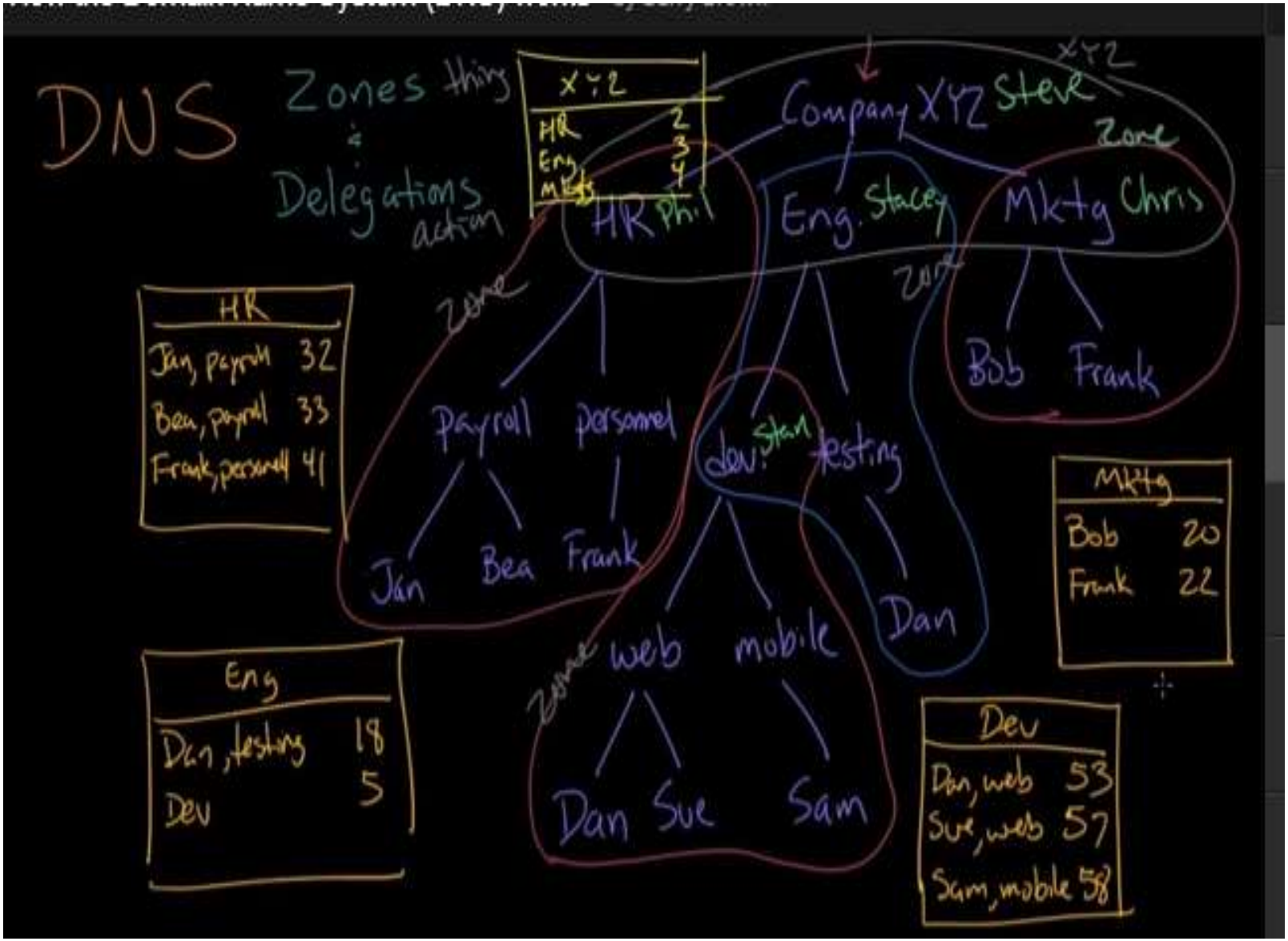
Domain

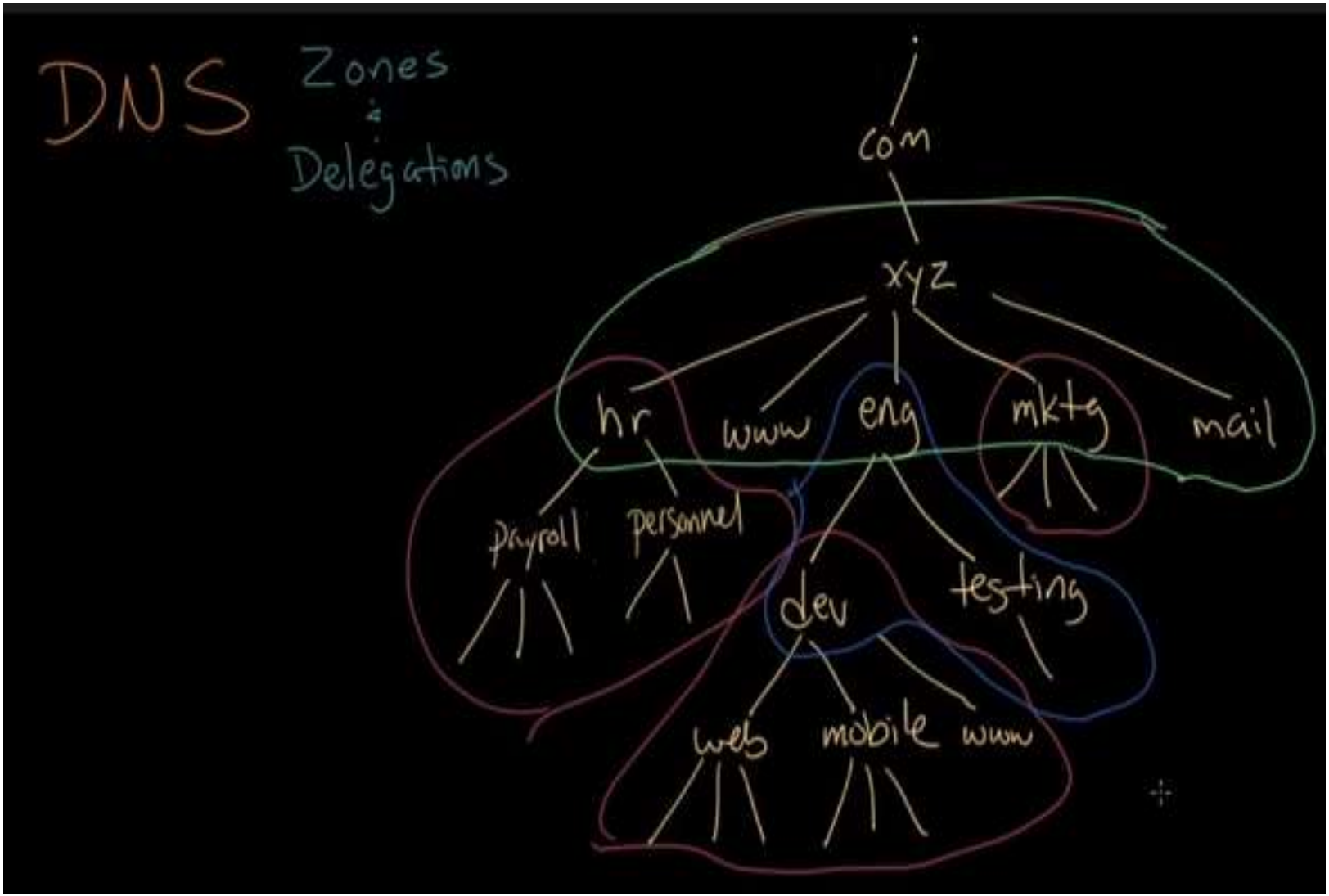
Subdomain



aspmx.l.google.com.

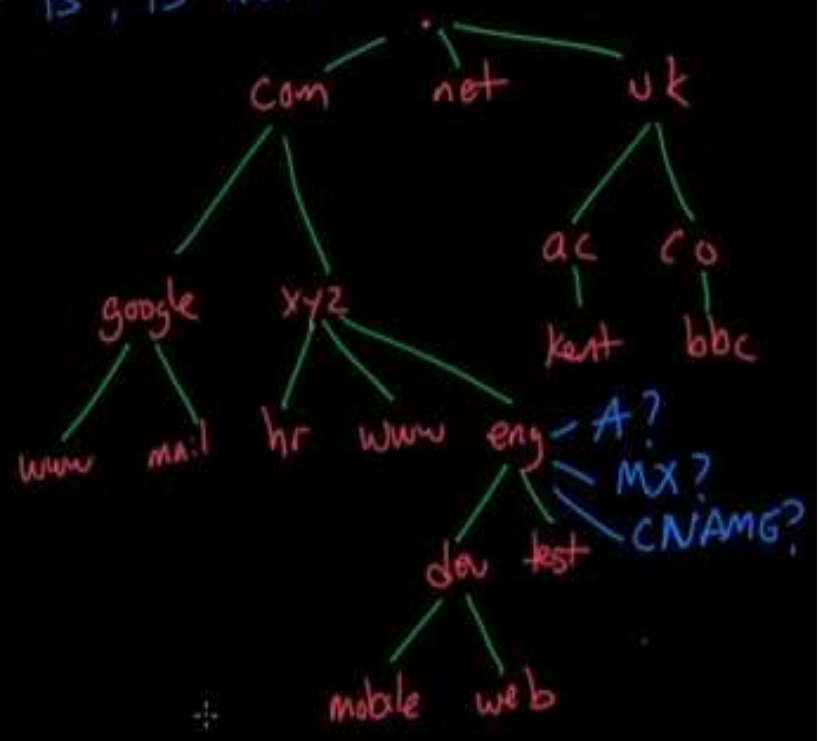

DNS
Zones
&
Delegations
XYZ
HR 2
Eng 3
Mktg 4
Company XYZ Steve
XYZ
Zone
HR Phil
Eng Stacey
Mktg Chris
Zone
Zone
HR
Jan, payroll 32
Bea, payroll 33
Frank, personell 41
payroll
personnel
dev Stan
testing
Bob
Frank
Jan
Bea
Frank
Dan
Mktg
Bob 20
Frank 22
Zone
web
mobile
Eng
Dan, testing 18
Dev 5
Dan
Sue
Sam
Dev
Dan, web 53
Sue, web 57
Sam, mobile 58

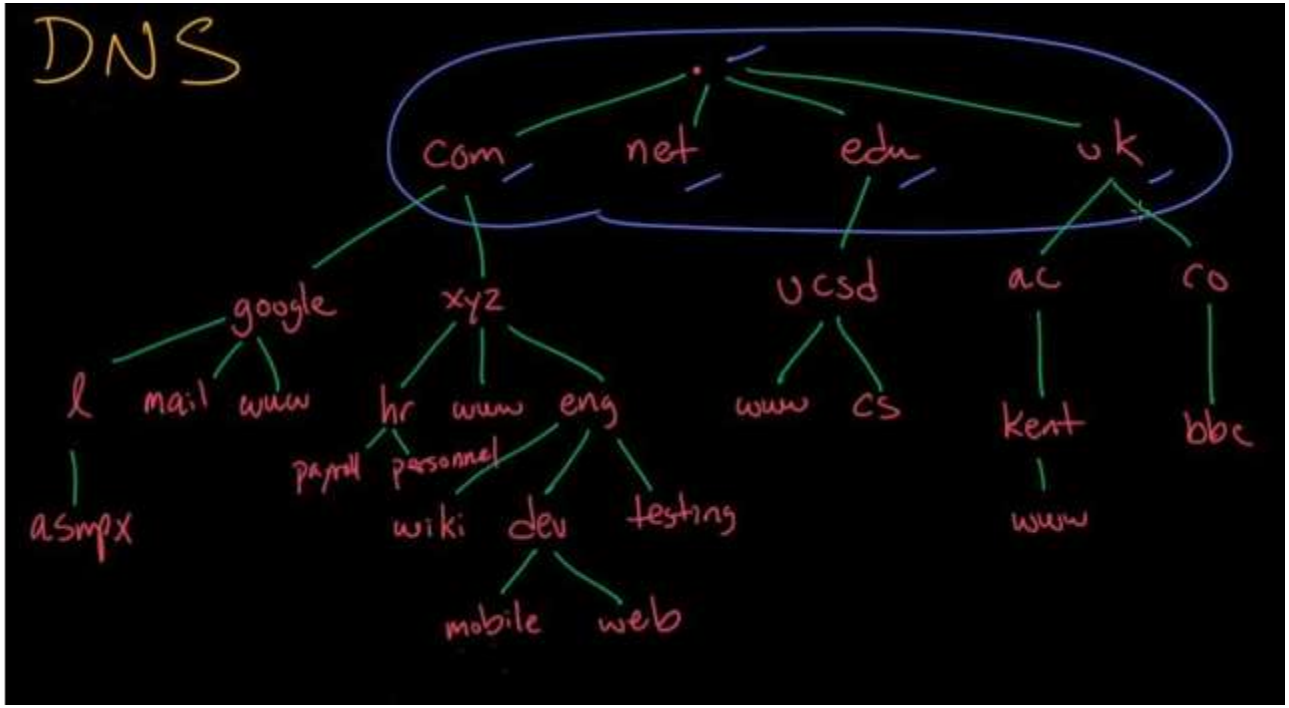

DNS
Zones thing
&
Delegations
action
XYZ
HR 2
Eng 3
Mktg 4
Company XYZ Steve
XYZ
Zone
HR Phil
Eng. Stacey
Mktg Chris
Zone
Zone
Bob Frank
Payroll
personnel
dev. Stan
testing
Jan Bea Frank
Dan
Zone
web
mobile
Dan Sue Sam
HR
Jan, payroll 32
Bea, payroll 33
Frank, personell 41
Mktg
Bob 20
Frank 22
Eng
Dan, testing 18
Dev 5
Dev
Dan, web 53
Sue, web 57
Sam, mobile 58

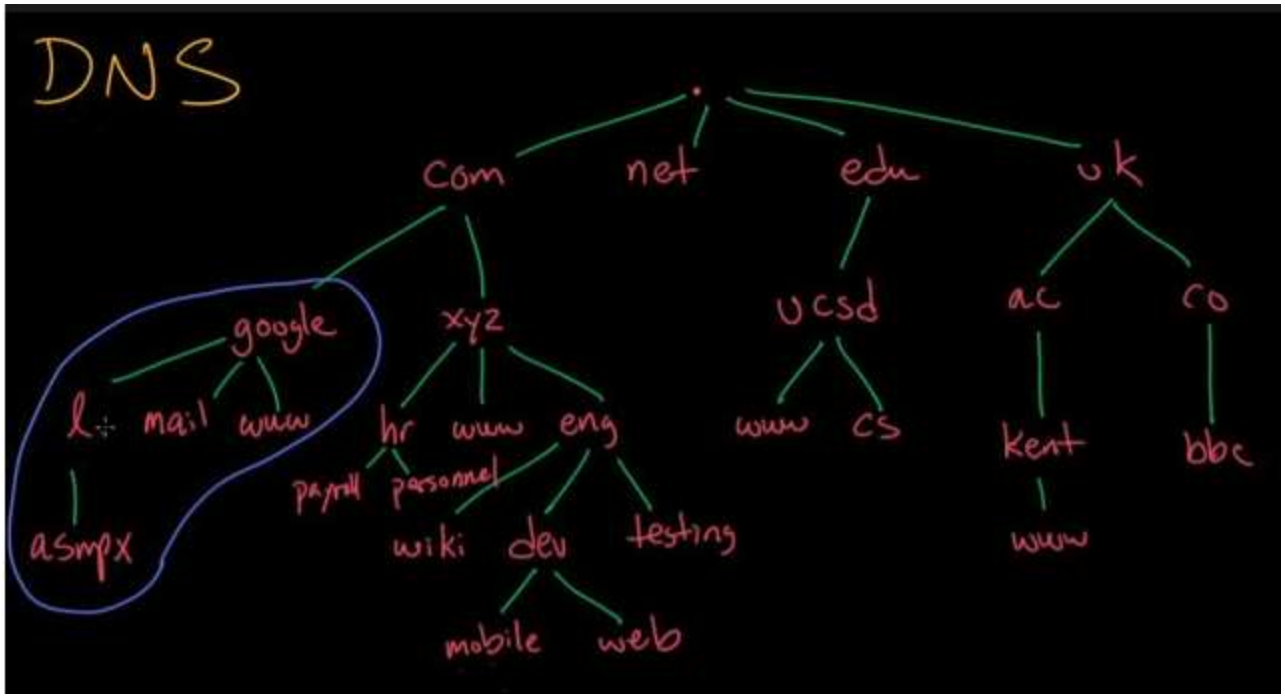

DNS
Zones
&
Delegations
com
xyz
hr
www
eng
mktg
mail
payroll
personnel
dev
testing
web
mobile
www

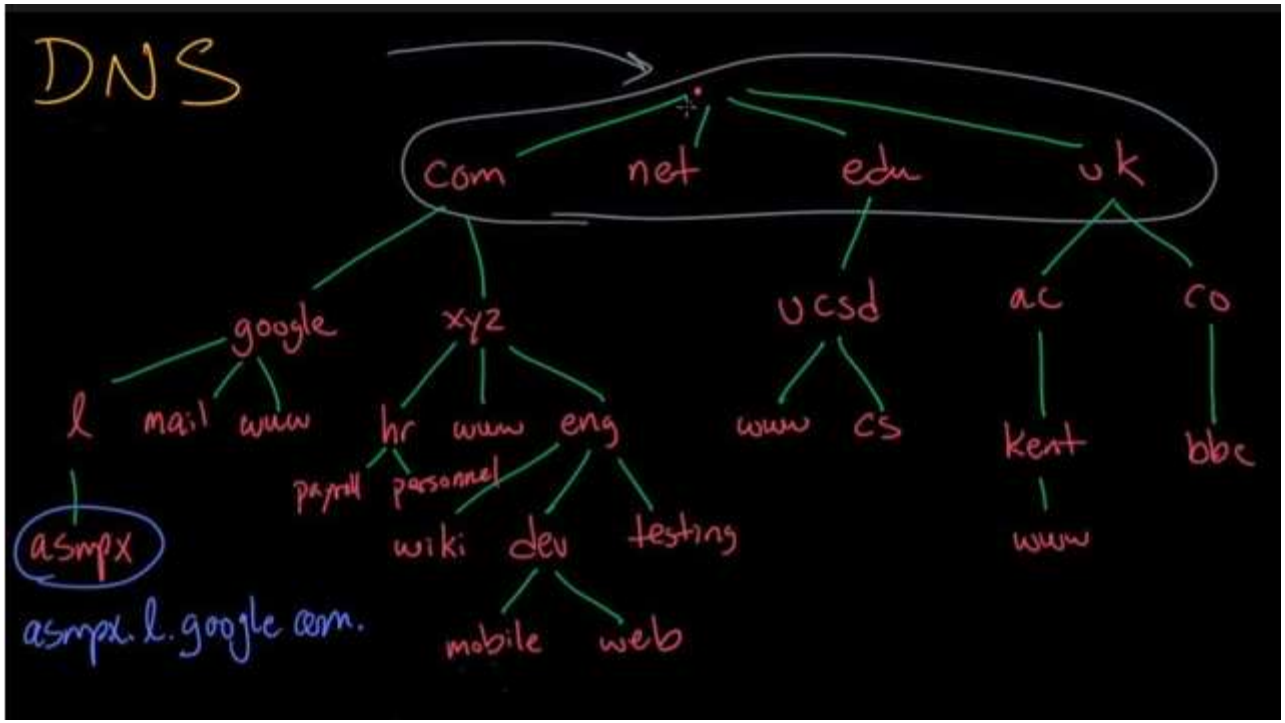
DNS What it is & is not.

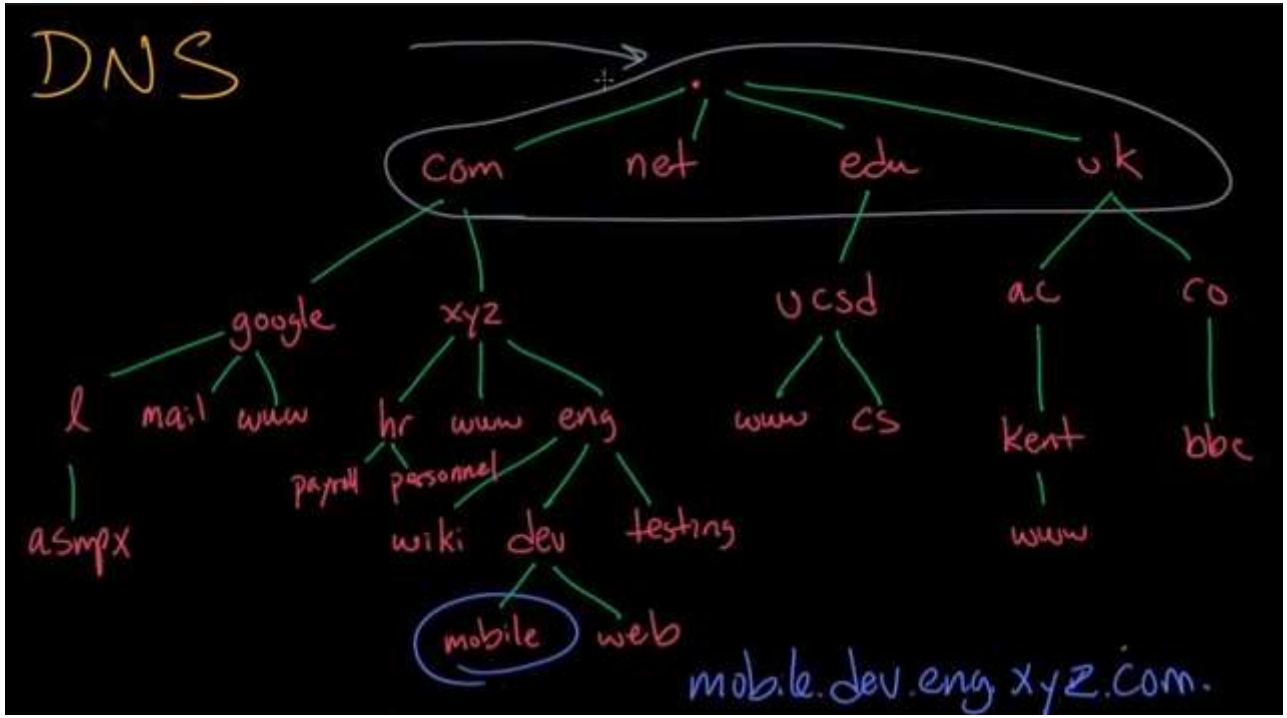
- A Database
- Not geography
- Not IP Routing

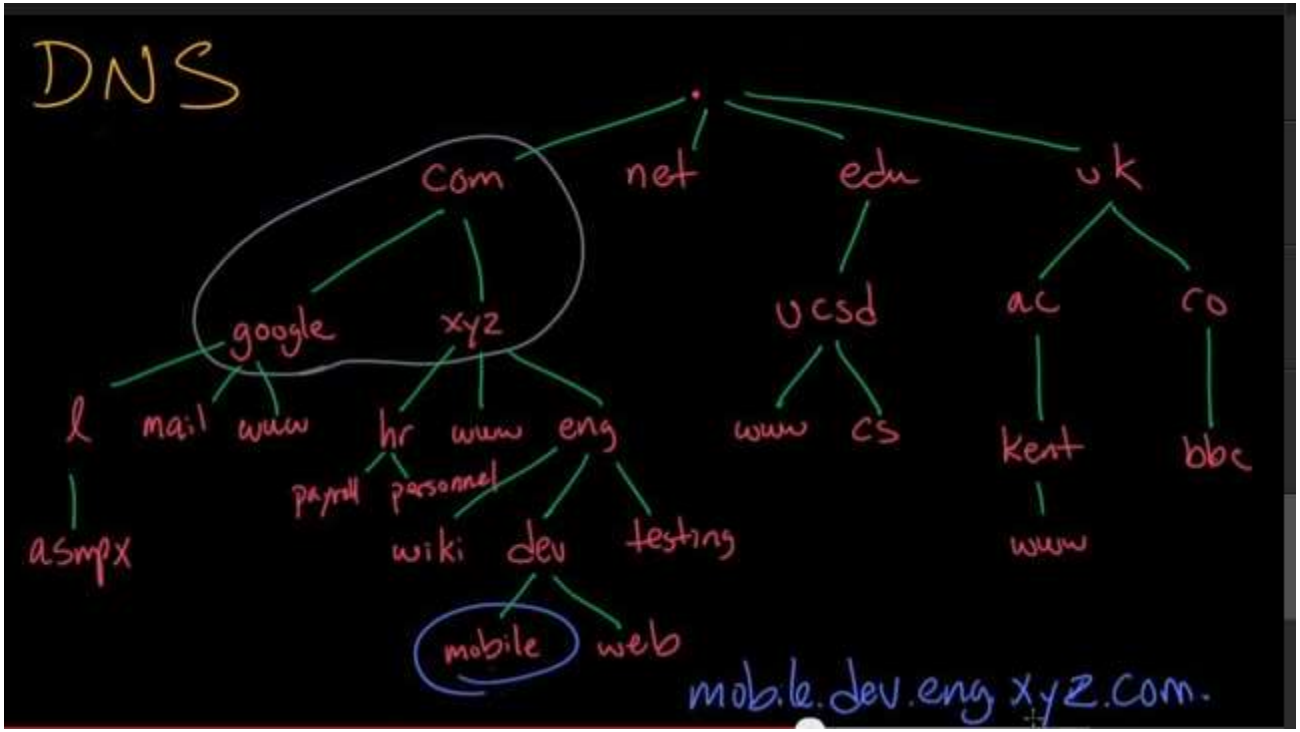


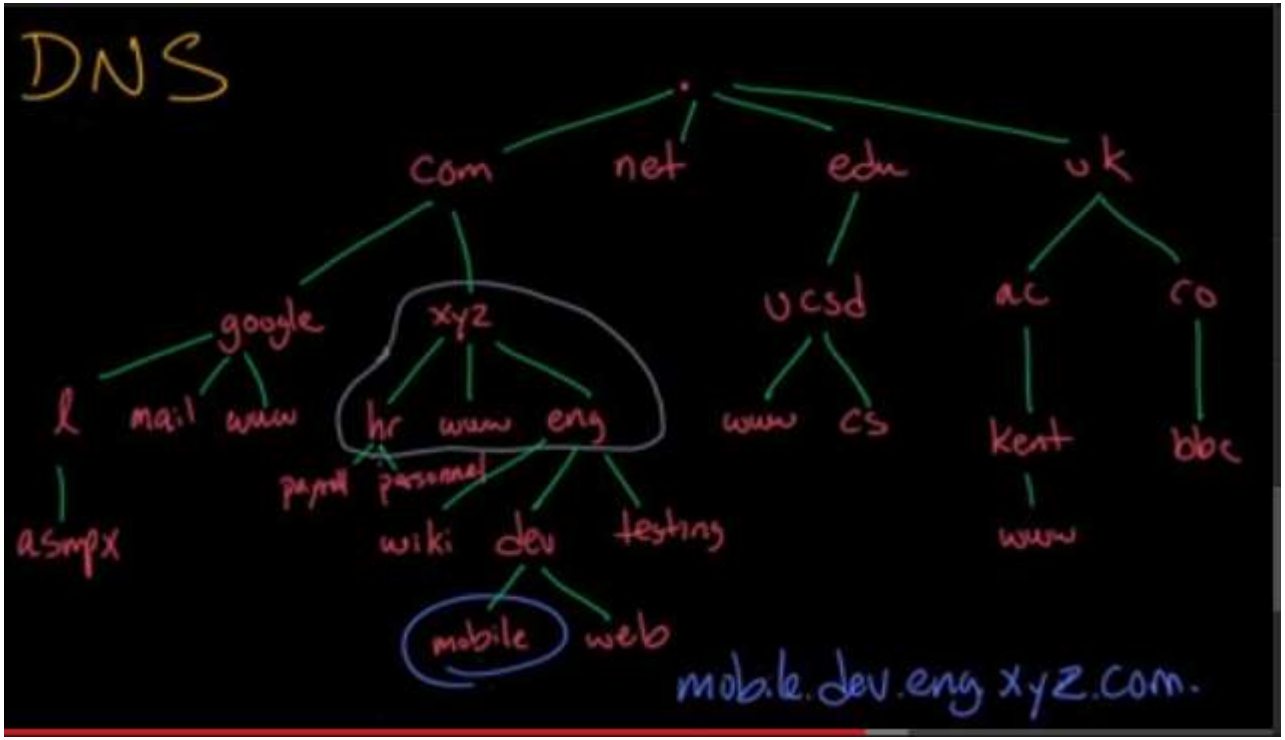

DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
asmpx
payroll
personnel
wiki
dev
testing
www
mobile
web

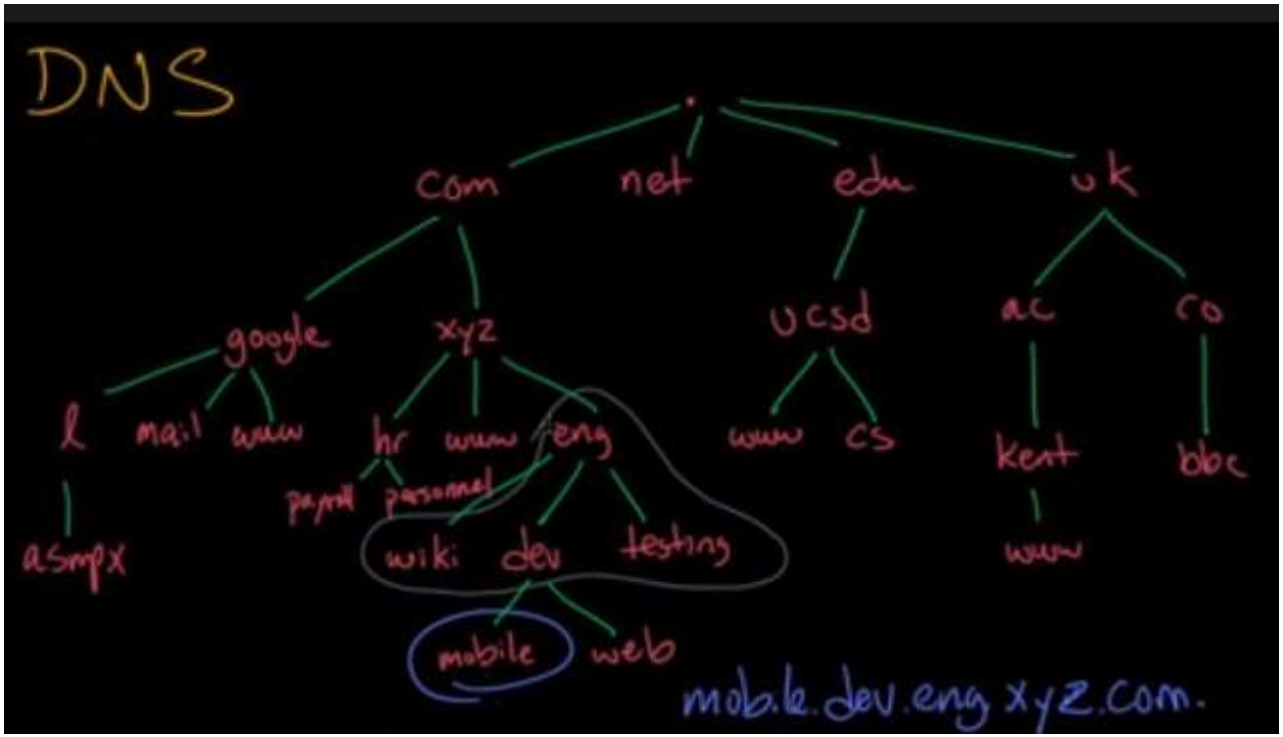

DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l.
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web

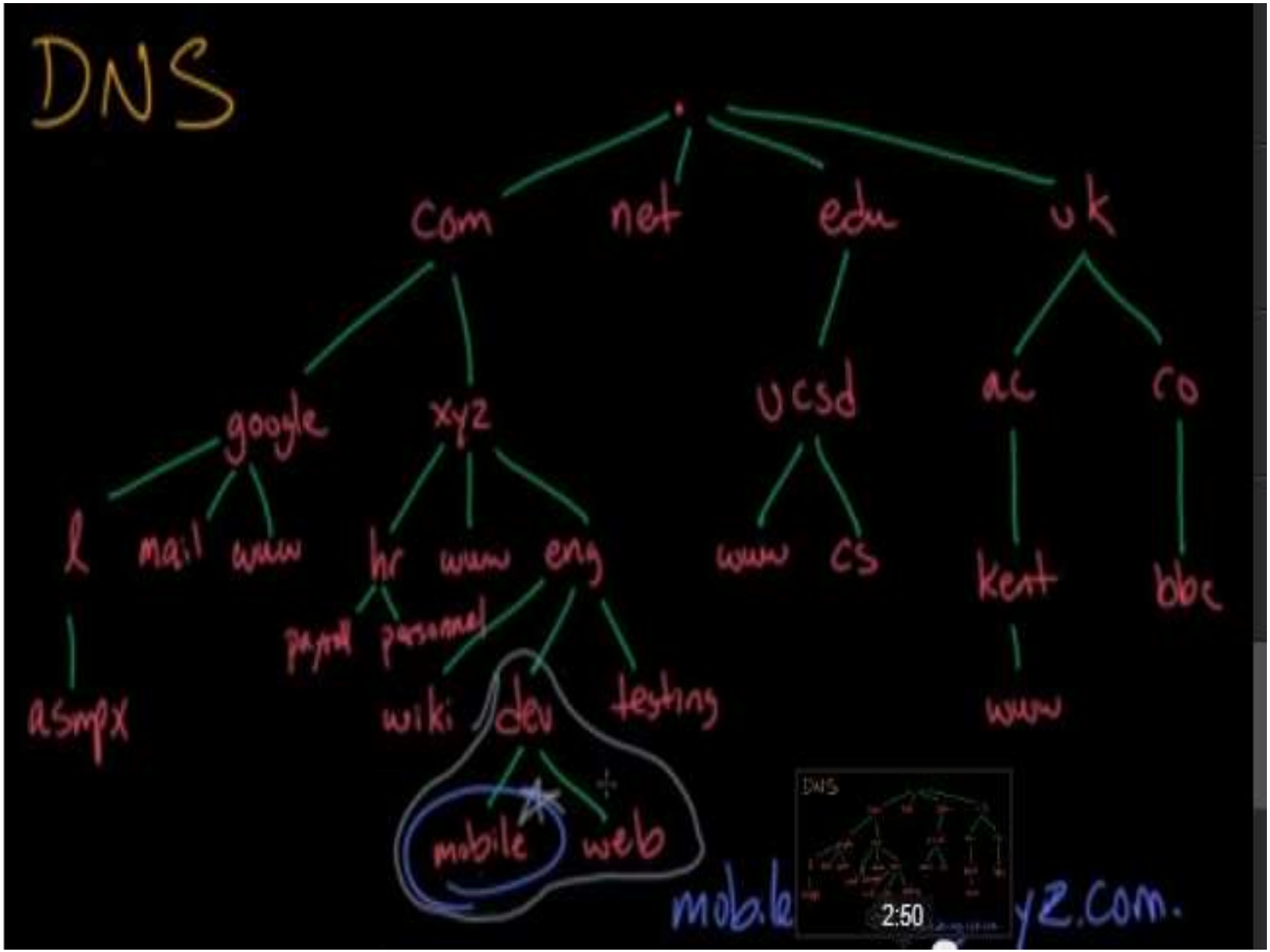

DNS
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
asmpx.l.google.com.
mobile
web


DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web
mobile.dev.eng.xyz.com.


DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web
mobile.dev.eng.xyz.com.


DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web
mobile.dev.eng.xyz.com.


DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web
mobile.dev.eng.xyz.com.


DNS
.
com
net
edu
uk
google
xyz
ucsd
ac
co
l
mail
www
hr
www
eng
www
cs
kent
bbc
payroll
personnel
asmpx
wiki
dev
testing
www
mobile
web
mobile
y2.com.
DNS
2:50

CNAME record

From Wikipedia, the free encyclopedia

A **CNAME record** is an abbreviation for **Canonical Name record** and is a type of resource record in the Domain Name System (DNS) that specifies that the domain name is an alias of another, canonical domain name. Here "canonical" usually means: a more generally accepted or standard name.

This is convenient when running multiple services (like an FTP server *and* a webserver; each running on different ports) from a single IP address. You can, for example, point `ftp.example.com` and `www.example.com` to the A record `example.com`, which in turn points to the IP-address. Then, if you ever need to change the IP-address, you only have to change it in one place. CNAME records must always be pointed to another domain name, never to an IP-address.